

HOSHIZAKI TEMP GUARD FH1-SSB REFRIGERATOR SERVICE MANUAL



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RFH1-SSB-HD

Refrigerator

Freezer (auxiliary code P5 and earlier)

Freezer (auxiliary code Q5 and later)

RFH-SSB(-HD)

RFH3-SSB(-HD)

PTR1SSB-xxxx, PTR1-SSB01-10

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SERVICE MANUAL



Number: 73077
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IMPORTANT

Only qualified service technicians should attempt to service or maintain this unit. No such service or maintenance should be undertaken until the technician has thoroughly read this Service Manual.

HOSHIZAKI provides this manual primarily to assist qualified service technicians in the service and maintenance of the unit.

Should the reader have any questions or concerns which have not been satisfactorily addressed, please call, write or send an e-mail message to the HOSHIZAKI Technical Support Department for assistance.

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Attn: HOSHIZAKI Technical Support Department

Phone: 1-800-233-1940 Technical Service
(770) 487-2331

Fax: 1-800-843-1056
(770) 487-3360

E-mail: techsupport@hoshizaki.com

Web Site: www.hoshizaki.com

NOTE: To expedite assistance, all correspondence/communication MUST include the following information:

- Model Number
- Serial Number
- Complete and detailed explanation of the problem

Please review this manual. It should be read carefully before the unit is serviced or maintenance operations are performed. Only qualified service technicians should service and maintain the unit. This manual should be made available to the technician prior to service or maintenance.

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I. Specifications

A. Nameplate Ratings

	AC Supply Voltage	Amperes	Design Pressure (PSIG)		Refrigerant
			HI	LO	R-404A
RH1-SSB / RH1-SSB-HD	115/60/1	7.0	450	200	12.2 OZ
RH1-SSB-GD	115/60/1	10.7	450	200	16.1 OZ
FH1-SSB / FH1-SSB-HD	115/60/1	12.0	450	250	15.2 OZ
RH2-SSB / RH2-SSB-HD	115/60/1	10.0	450	200	18.2 OZ
RH2-SSB-GD	115/60/1	11.8	400	200	21.5 OZ
FH2-SSB / FH2-SSB-HD	115/60/1	15.5	450	250	20.1 OZ
RH3-SSB / RH3-SSB-HD	115/60/1	13.0	450	200	23.2 OZ
RH3-SSB-GD	115/60/1	15.5	450	250	31.4 OZ
RIR2-SSB51-02	115/60/1	11.0	450	200	18.2 OZ
PTR1SSB-xxxx / PTR1-SSB01-10	115/60/1	12.0	450	250	15.5 OZ

	AC Supply Voltage	Amperes	Design Pressure (PSIG)				Refrigerant	
			Refrigerator		Freezer		R-404A	
			HI	LO	HI	LO	Refrigerator	Freezer
RFH2-SSB / RFH2-SSB-HD	115/60/1	19	450	200	450	250	12.2 OZ	15.2 OZ
RFH3-SSB / RFH3-SSB-HD	115/60/1	22	450	200	450	250	18.2 OZ	15.2 OZ
RFH1-SSB-HD	115/60/1	16.0	450	250	450	250	9.5 OZ	12.0 OZ

See the nameplate for electrical and refrigeration specifications. The nameplate is located on the right side wall of the cabinet interior.

Note: We reserve the right to make changes in specifications and design without prior notice.

B. Dimensions

1. Notes for All Units

- a) Units: mm (in.)
- b) Legs have 25.4 mm (1 in.) height adjustment

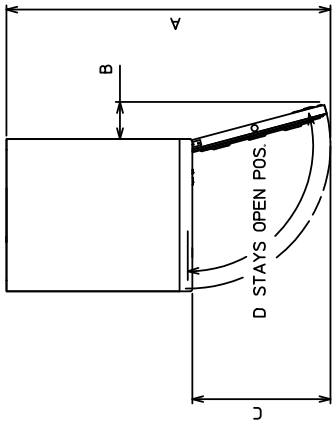
	Door Opening		Total Refrigerated Volume ft³		Total Shelf Space ft²	
	Width mm (in.)	Height mm (in.)				
RH1-SSB / FH1-SSB	554.8 (21.8)	1507(59.3)	22.3		11.5	
RH1-SSB-GD	554.8 (21.8)	1507(59.3)	22.3			
RH1-SSB-HD / FH1-SSB-HD	554.8 (21.8)	681.7(26.8)	22.3			
RH2-SSB / FH2-SSB	554.8 (21.8)	1507(59.3)	48.3		25.3	
RH2-SSB-GD / RIR2-SSB51-02	554.8 (21.8)	1507(59.3)	48.3			
RH2-SSB-HD / FH2-SSB-HD	554.8 (21.8)	681.7(26.8)	48.3			
RH3-SSB / FH3-SSB	554.8 (21.8)	1507(59.3)	73.7		39	
RH3-SSB-GD	554.8 (21.8)	1507(59.3)	73.7			
RH3-SSB-HD / FH3-SSB-HD	554.8 (21.8)	681.7(26.8)	73.7			
			Refrigerator	Freezer	Refrigerator	Freezer
RFH2-SSB	554.8 (21.8)	1507(59.3)	22.3	22.3	11.5	11.5
RFH2-SSB-HD	554.8 (21.8)	681.7(26.8)	22.3	22.3	11.5	11.5
RFH3-SSB	554.8 (21.8)	1507(59.3)	48.3	22.3	25.3	11.5
RFH3-SSB-HD	554.8 (21.8)	681.7(26.8)	48.3	22.3	25.3	11.5
RFH1-SSB-HD	554.8 (21.8)	681.7 (26.8)	9.1	10.5	11.5 (total)	

	Depth with Doors Open	Door Opening		Total Refrigerated Volume ft³	Total Shelf Space ft²
		Width mm (in.)	Height mm (in.)		
PTR1SSB-FSFS	2,177 (85.9)	554.8 (21.8)	1507(59.3)	23.8	13.6
PTR1SSB-FGFG	2,146.2 (84.7)				
PTR1SSB-HSHS	2,177 (85.9)	554.8 (21.8)	681.7 (26.8)		
PTR1SSB-HGHG	2,146.2 (84.7)				
PTR1SSB-HGHS / PTR1-SSB01-10 (Half glass and half solid door combo)	2,161.6 (85.3)				

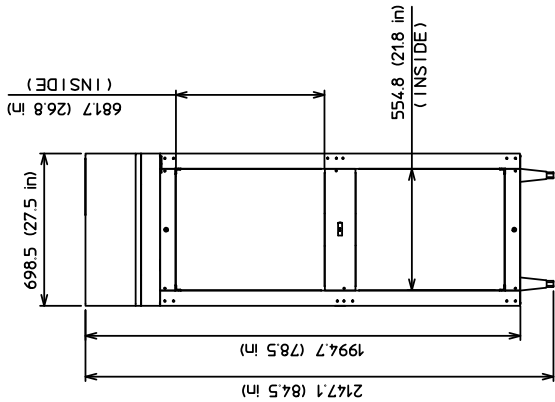
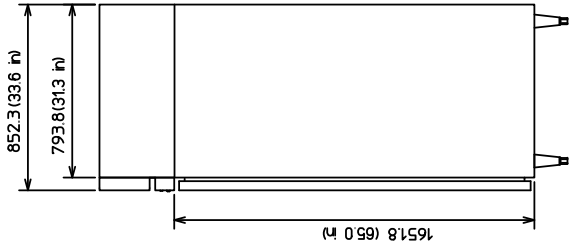
See the nameplate for electrical and refrigeration specifications. The nameplate is located on the right side wall of the cabinet interior.

Note: We reserve the right to make changes in specifications and design without prior notice.

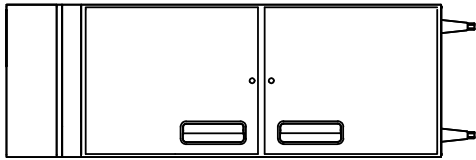
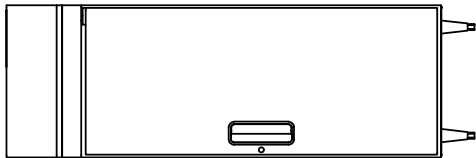
2. RH1-SSB(-HD), FH1-SSB(-HD), RFH1-SSB-HD, RH1-SSB-GD



MODELS	A	B	C	D
*H1-SSB,	1485.4 (58.5 in)	170 (6.7 in)	633.1 (25 in)	105°
*H1-SSB-HD				
*H1-SSB-GD	1470 (57.9 in)	0 (0 in)	617.7 (24.4 in)	90°



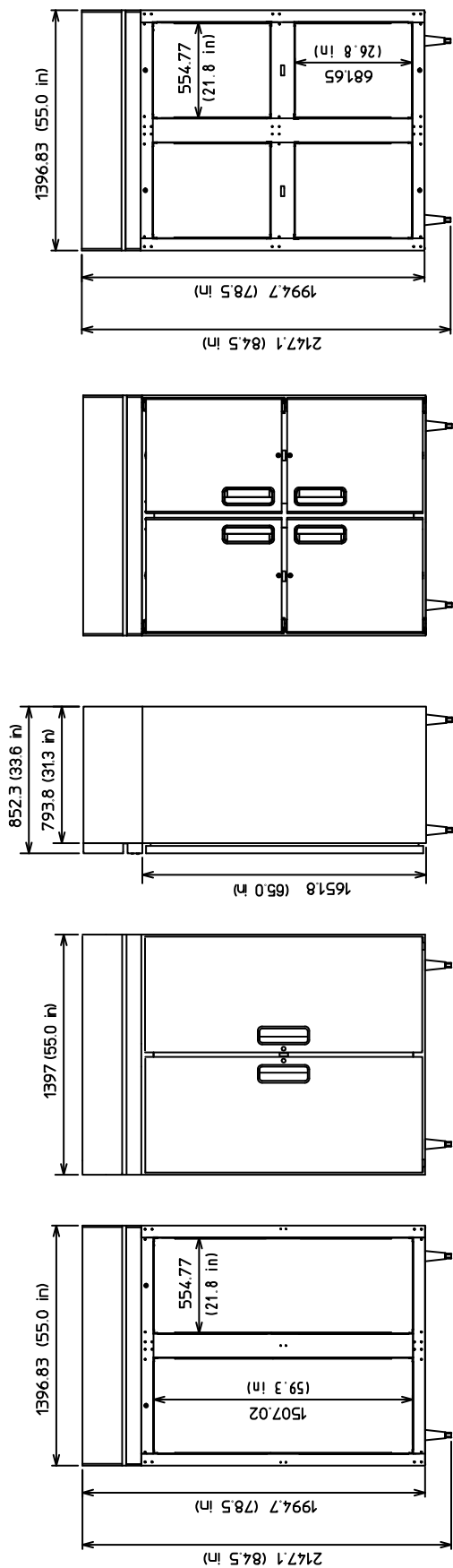
THESE VIEWS FOR *H1-SSB AND *H1-SSB-GD



THESE VIEWS FOR *H1-SSB-HD

mm (in.)

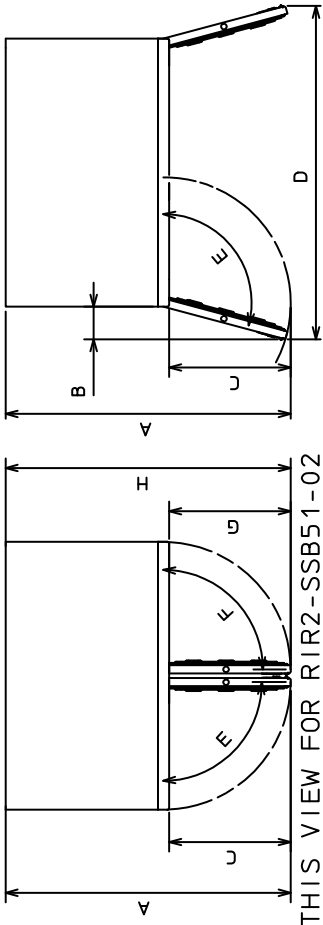
3. RH2-SSB(-HD), FH2-SSB(-HD), RFH2-SSB(-HD), RH2-SSB-GD



THESE VIEWS FOR *H2-SSB-HD

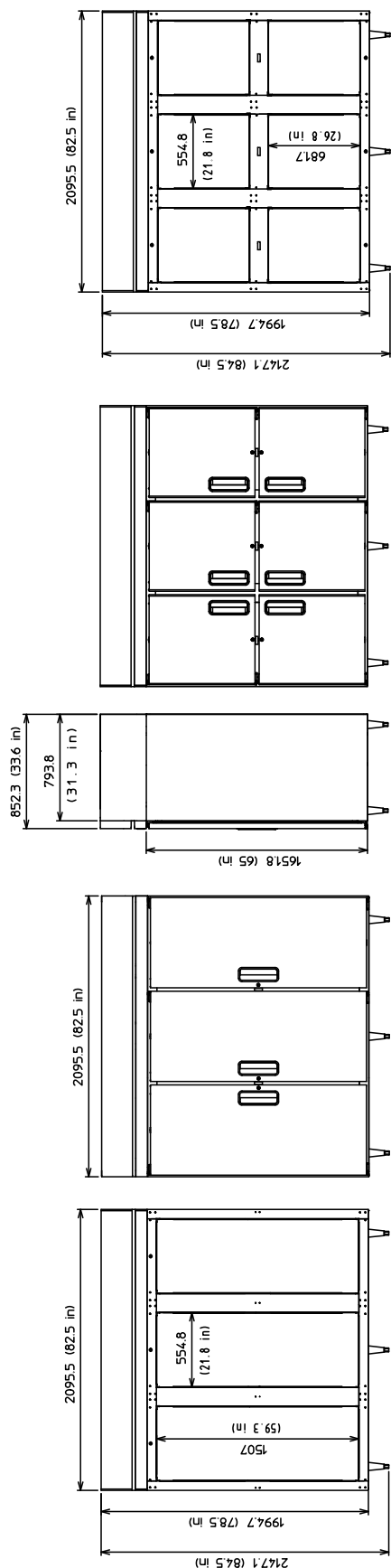
THESE VIEWS FOR *H2-SSB & RIR2-SSB51-02

MODELS	A	B	C	D	STAY OPEN POS.		G	H
					E	F		
*H2-SSB, *H2-SSB-HD	1485.4 (58.5 in)	170 (6.7 in)	633.1 (25 in)	1737 (68.4 in)	105°	NA	NA	NA
*H2-SSB-GD	1470 (57.9 in)	0	617.7 (24.4 in)	1361.9 (53.6 in)	90°	NA	NA	NA
RIR2-SSB51-02	1470 (57.9 in)	NA	617.7 (24.4 in)	NA	90°	105°	633.1 (25 in)	1485.4 (58.5 in)



mm (in.)

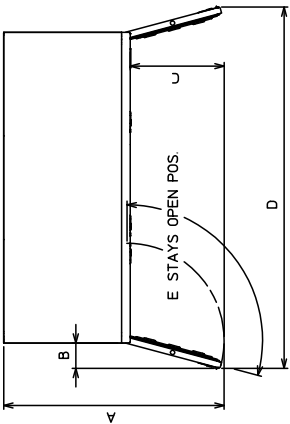
4. RH3-SSB(-HD), FH3-SSB(-HD), RFH3-SSB(-HD), RH3-SSB-GD



THESE VIEWS FOR *H3-SSB-HD

THESE VIEWS FOR *H3-SSB

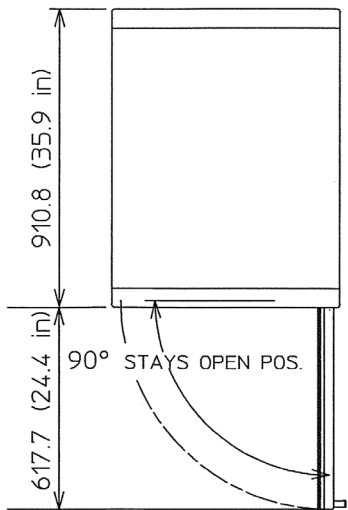
MODELS	A	B	C	D	E
*H3-SSB ,	1485.4 (58.5 in)	170 (6.7 in)	633.1 (25 in)	2435.4 (96.9 in)	105°
*H3-SSB-HD	1430 (57.9 in)	0	647.7 (24.4 in)	2059.7 (81.1 in)	90°



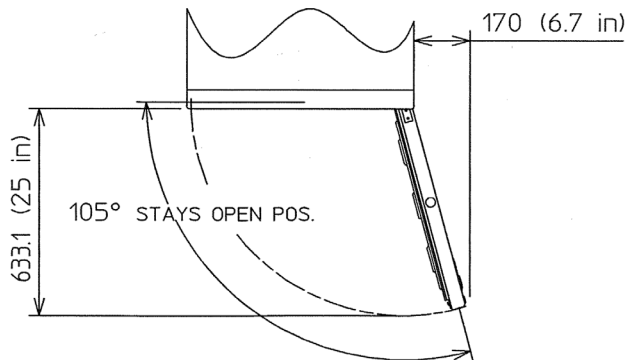
mm (in.)

5. PTR1SSB-xxxx, PTR1-SSB01-10

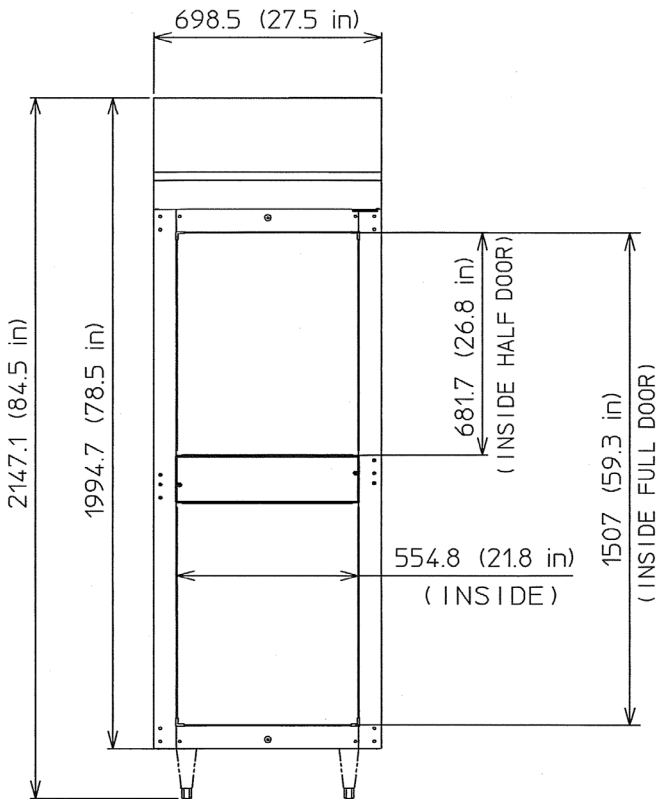
mm (in.)



DOOR SWING
GLASS DOOR



DOOR SWING
SOLID DOOR

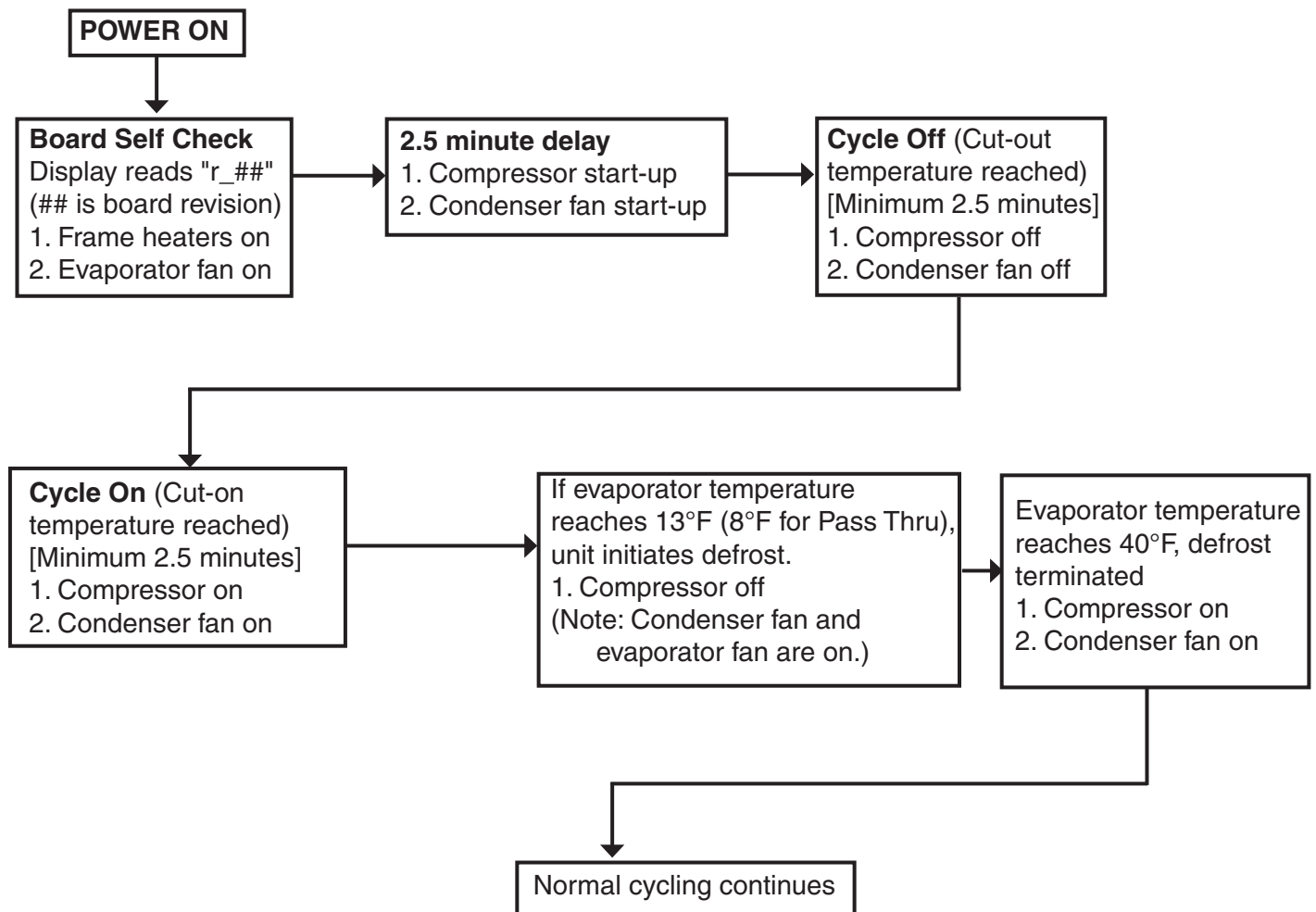


II. General Information

A. Sequence of Operation and Timing Charts

1. Refrigerator and Refrigerator Side of Dual Temp Models

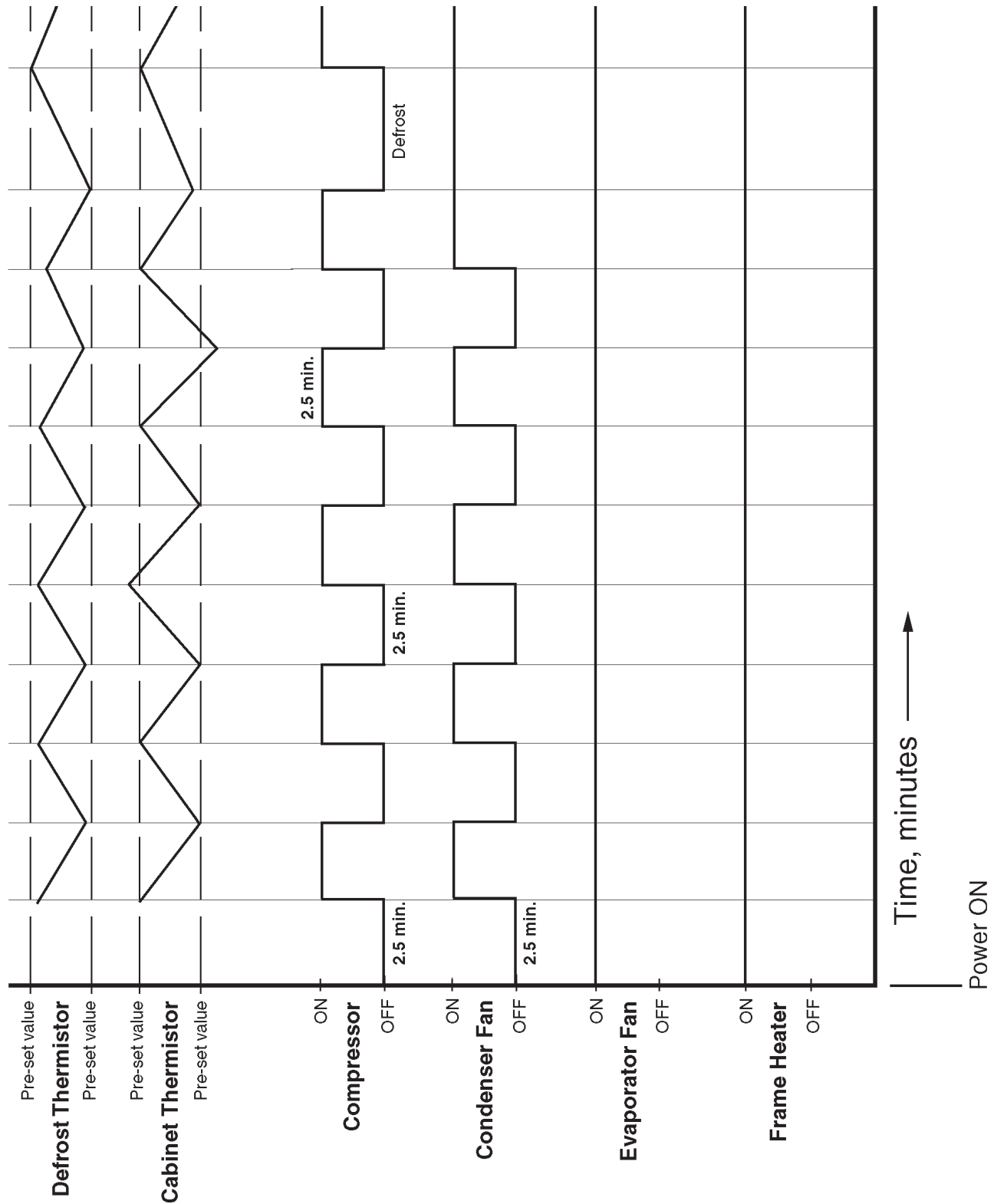
a) Sequence of Operation



Note: The start circuit of the compressor is timed such that at power-up and during any compressor off time, there will be at least a 2 1/2 minute delay before the compressor will start. The only exception is when the overload activates and deactivates. The compressor has a 2 1/2 minute minimum run time during every run cycle.

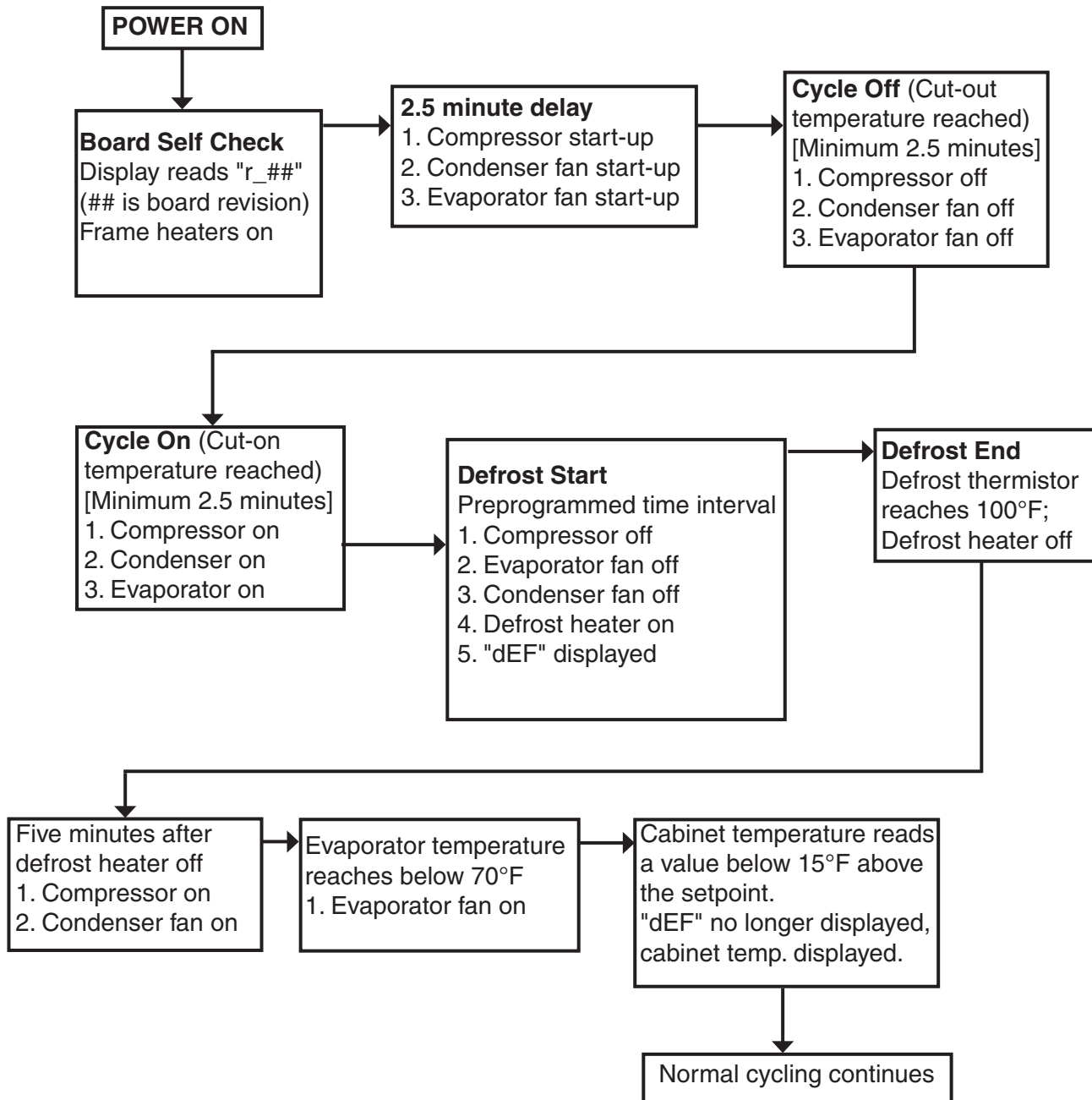
b) Timing Chart

For refrigerator, and refrigerator side of Dual Temp models.



2. Freezer (except freezer side of RFH1-SSB-HD) auxiliary code P5 and earlier

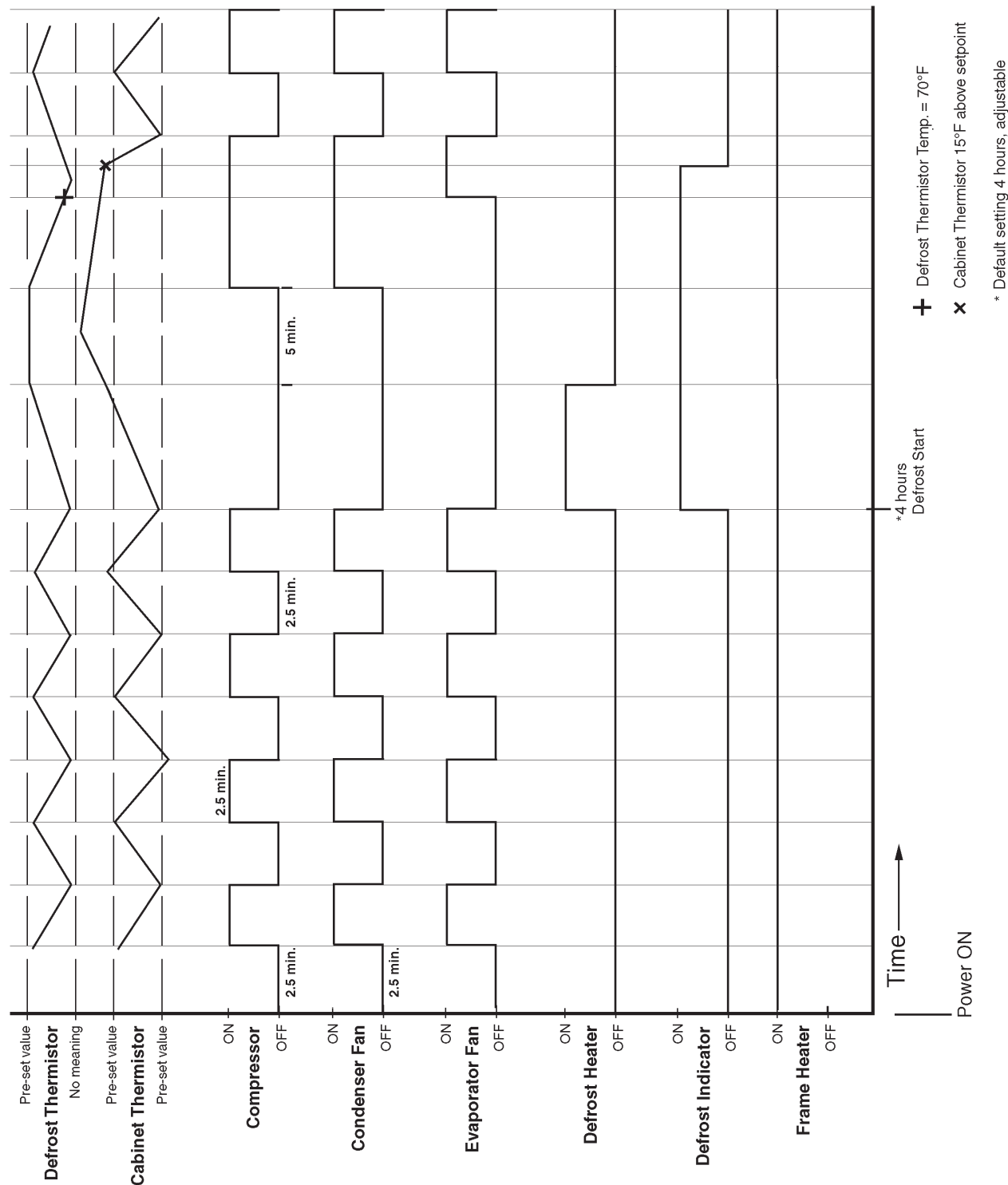
a) Sequence of Operation



Note: The start circuit of the compressor is timed such that at power-up and during any compressor off time, there will be at least a 2 1/2 minute delay before the compressor will start. The only exception is when the overload activates and deactivates. The compressor has a 2 1/2 minute minimum run time during every run cycle.

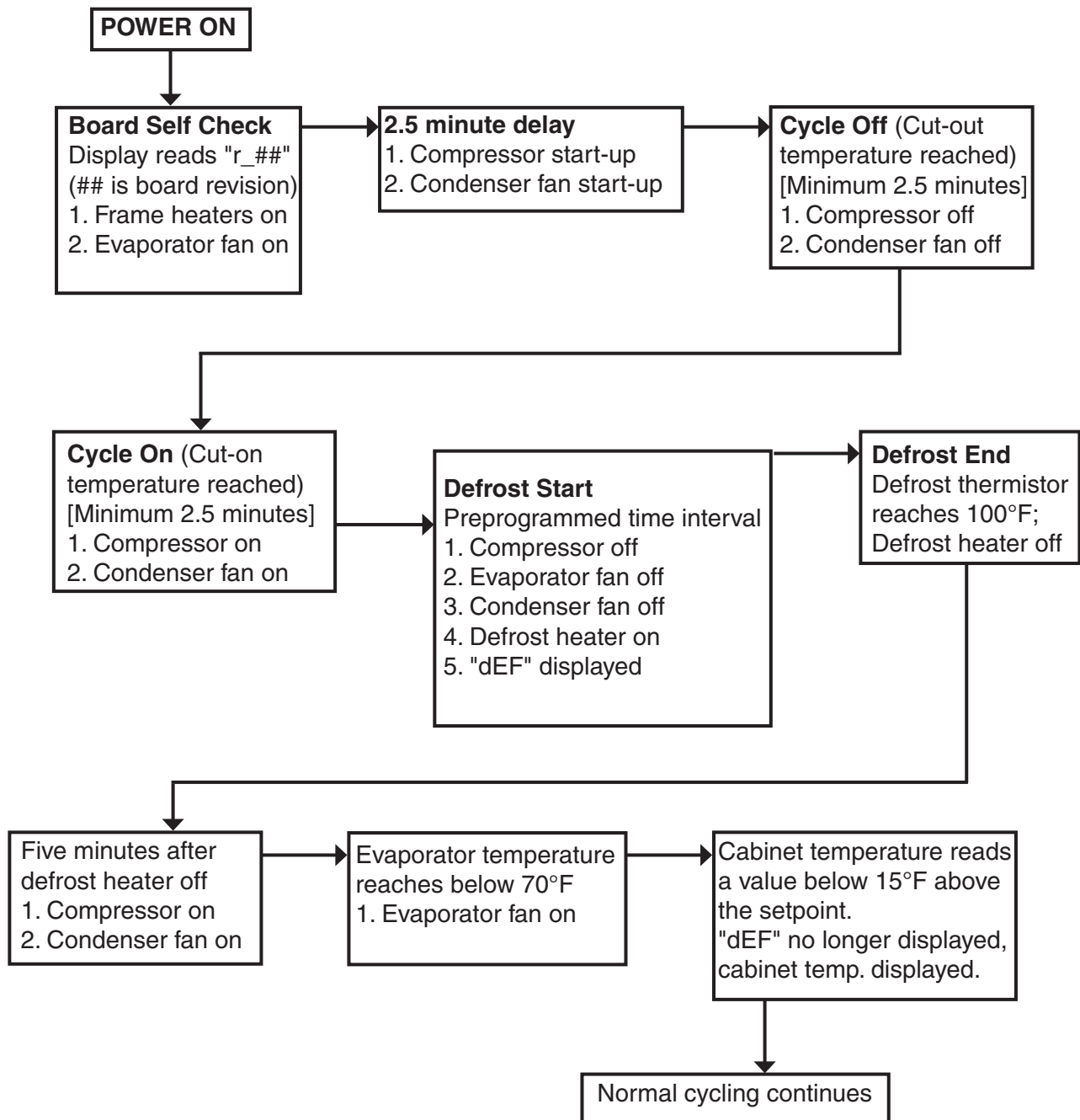
b) Timing Chart

For freezer (except freezer side of RFH1-SSB-HD) auxiliary code P5 and earlier.



3. Freezer (except freezer side of RFH1-SSB-HD) auxiliary code P6 and later

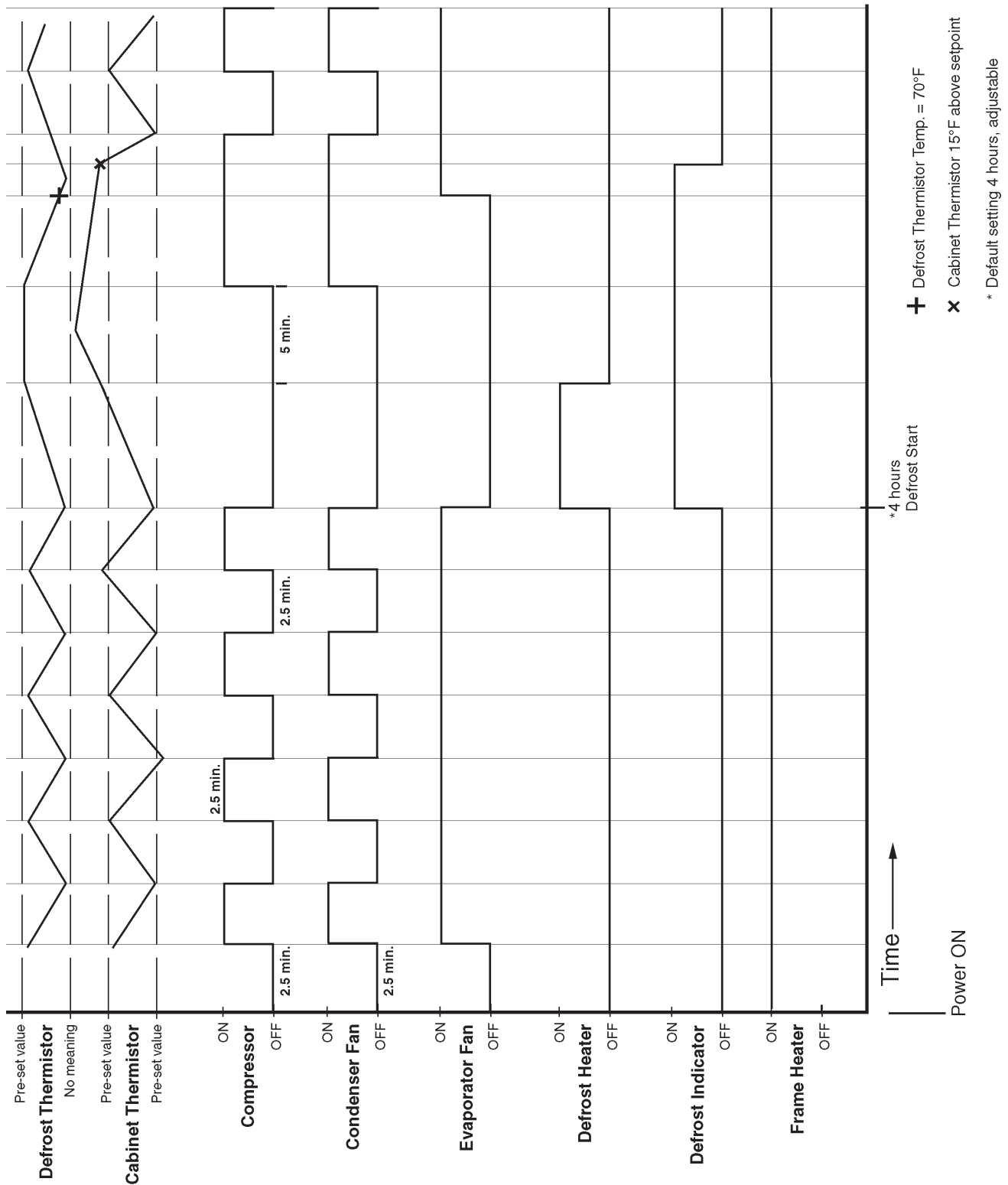
a) Sequence of Operation



Note: The start circuit of the compressor is timed such that at power-up and during any compressor off time, there will be at least a 2 1/2 minute delay before the compressor will start. The only exception is when the overload activates and deactivates. The compressor has a 2 1/2 minute minimum run time during every run cycle.

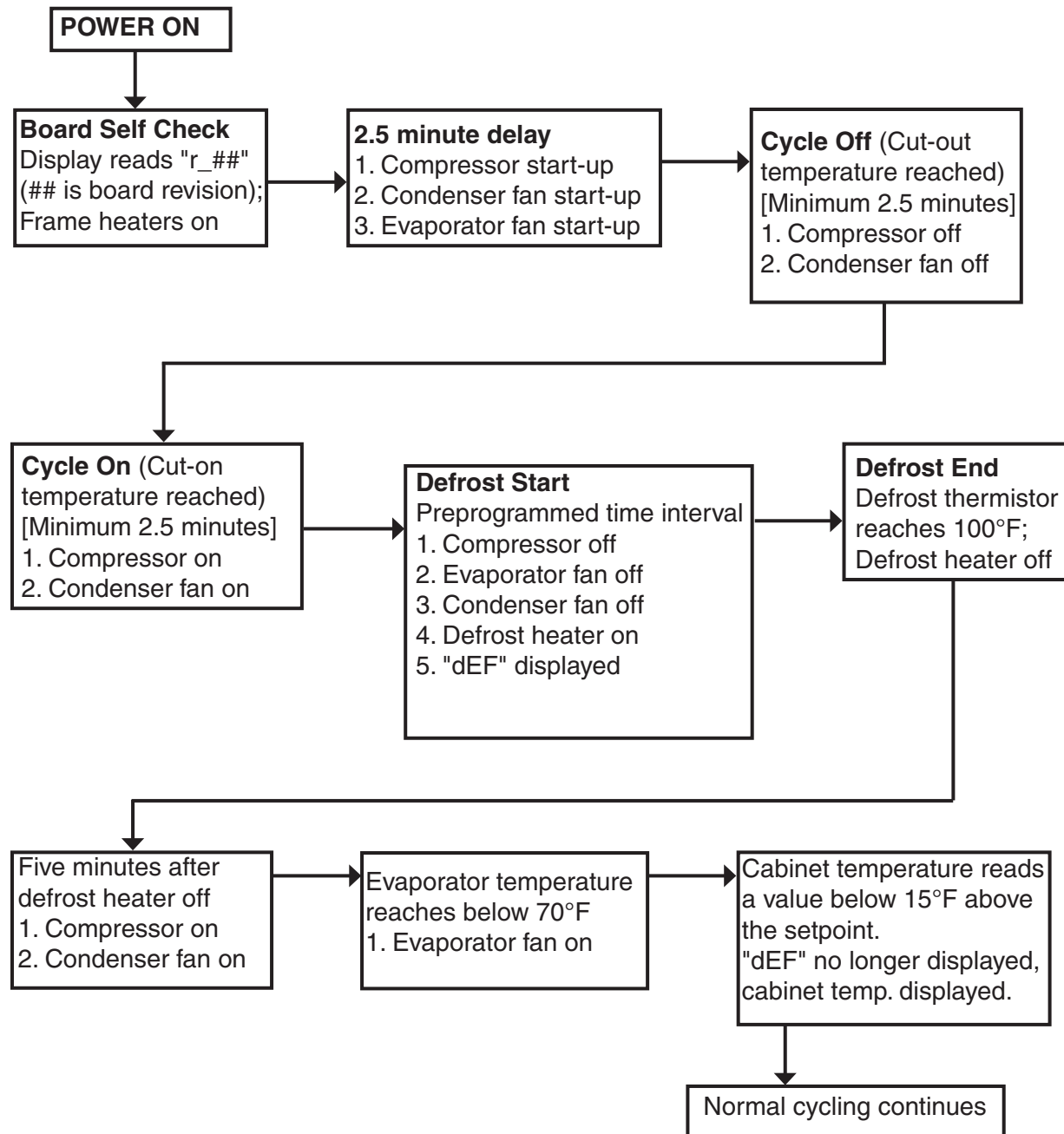
b) Timing Chart

For freezer (except freezer side of RFH1-SSB-HD) auxiliary code P6 and later.



4. Freezer Side of RFH1-SSB-HD

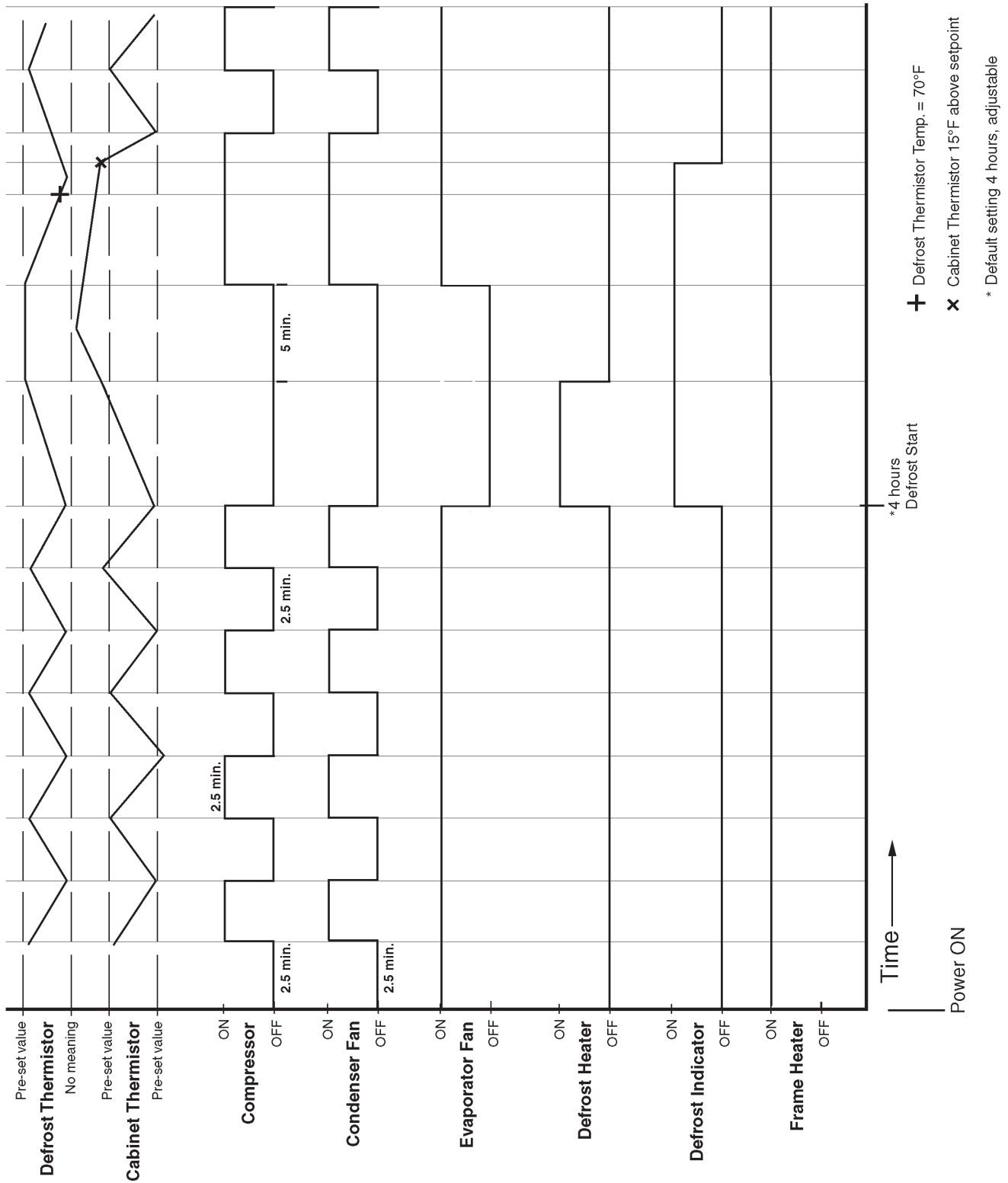
a) Sequence of Operation



Note: The start circuit of the compressor is timed such that at power-up and during any compressor off time, there will be at least a 2 1/2 minute delay before the compressor will start. The only exception is when the overload activates and deactivates. The compressor has a 2 1/2 minute minimum run time during every run cycle.

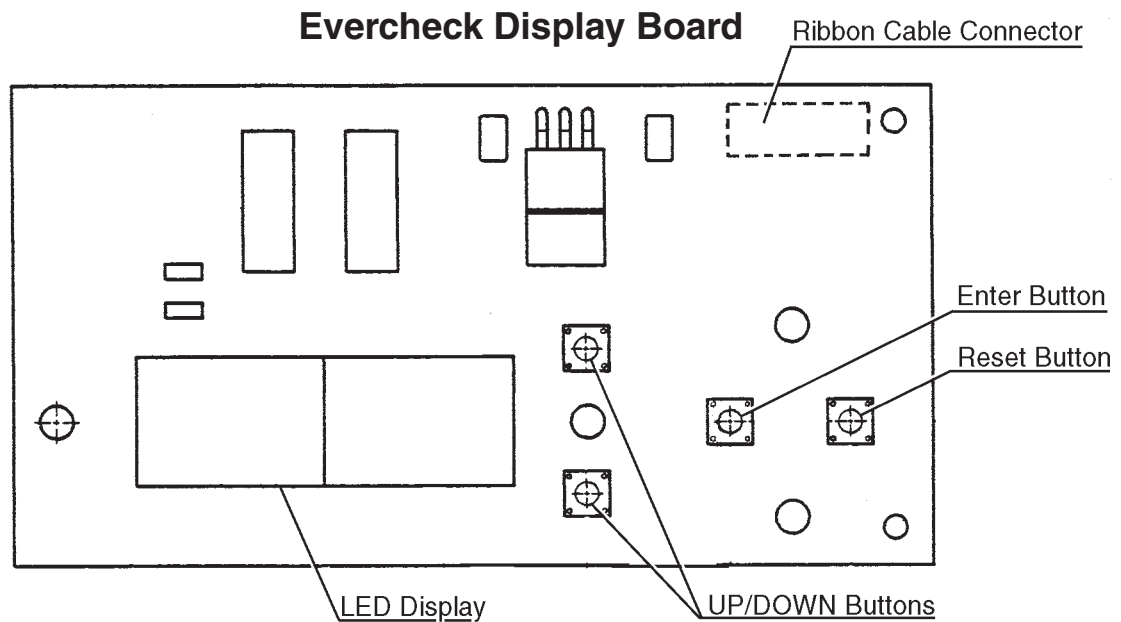
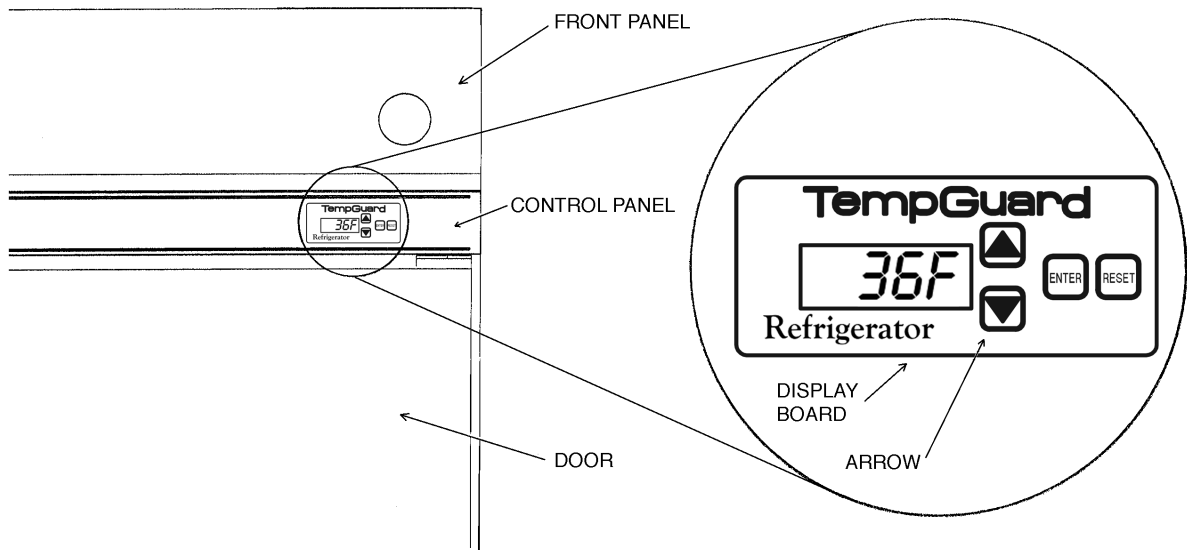
b) Timing Chart

For freezer side of RFH1-SSB-HD.



B. Display Board

The display board is located on the right hand side of the control panel between the top of the door and the front panel. From this area, information regarding the functioning of the unit can be obtained. When the unit is turned on, the first display reads "r_##", where "##" indicates the board revision level. From the display board, various electronic controls and functions can also be adjusted. See "D. Settings and Diagnostics."



C. Control Board

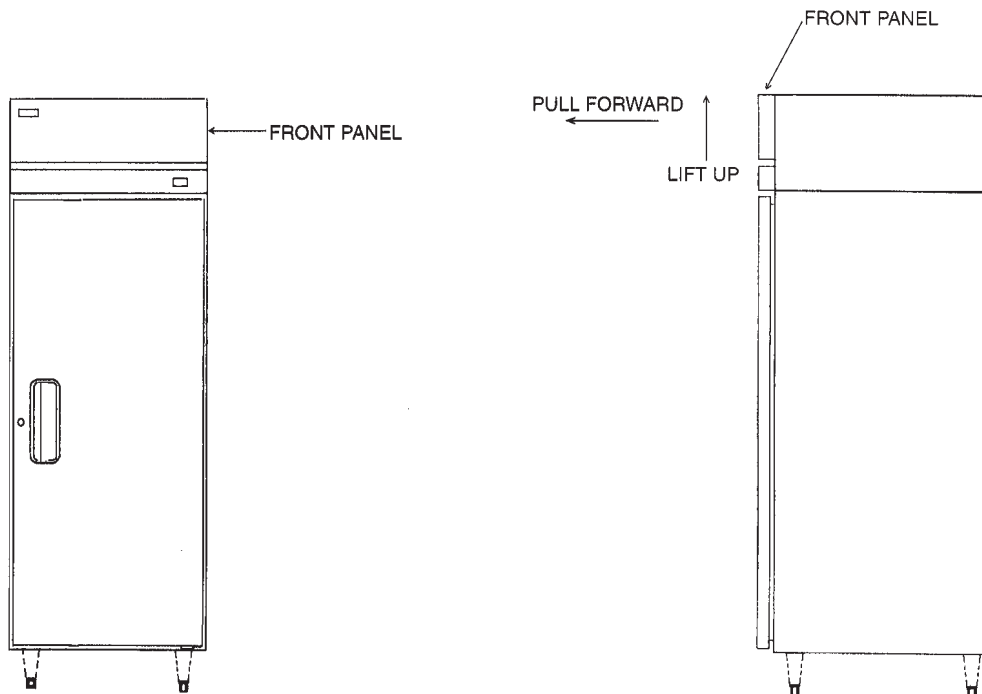
- A HOSHIZAKI exclusive solid-state control is employed in all Hoshizaki refrigerators and freezers.
- All models are pretested and factory-adjusted.

CAUTION

1. Fragile, handle very carefully.
2. The control board contains integrated circuits, which are susceptible to failure due to static discharge. It is especially important to touch the metal part of the unit before handling or replacing the board.
3. To prevent damage to the board, do not touch the electronic devices on the board or the back of the board.
4. Do not fix the electronic devices or parts on the board in the field. Always replace the whole board assembly if it goes bad.
5. Do not short out power supply to test for voltage.
6. Keep the thermistors, clogged filter leads, door switch leads, pressure switch leads and ribbon cable at least 1 1/2" away from high voltage leads (AC 100 volts or more) to protect against electrical noise.

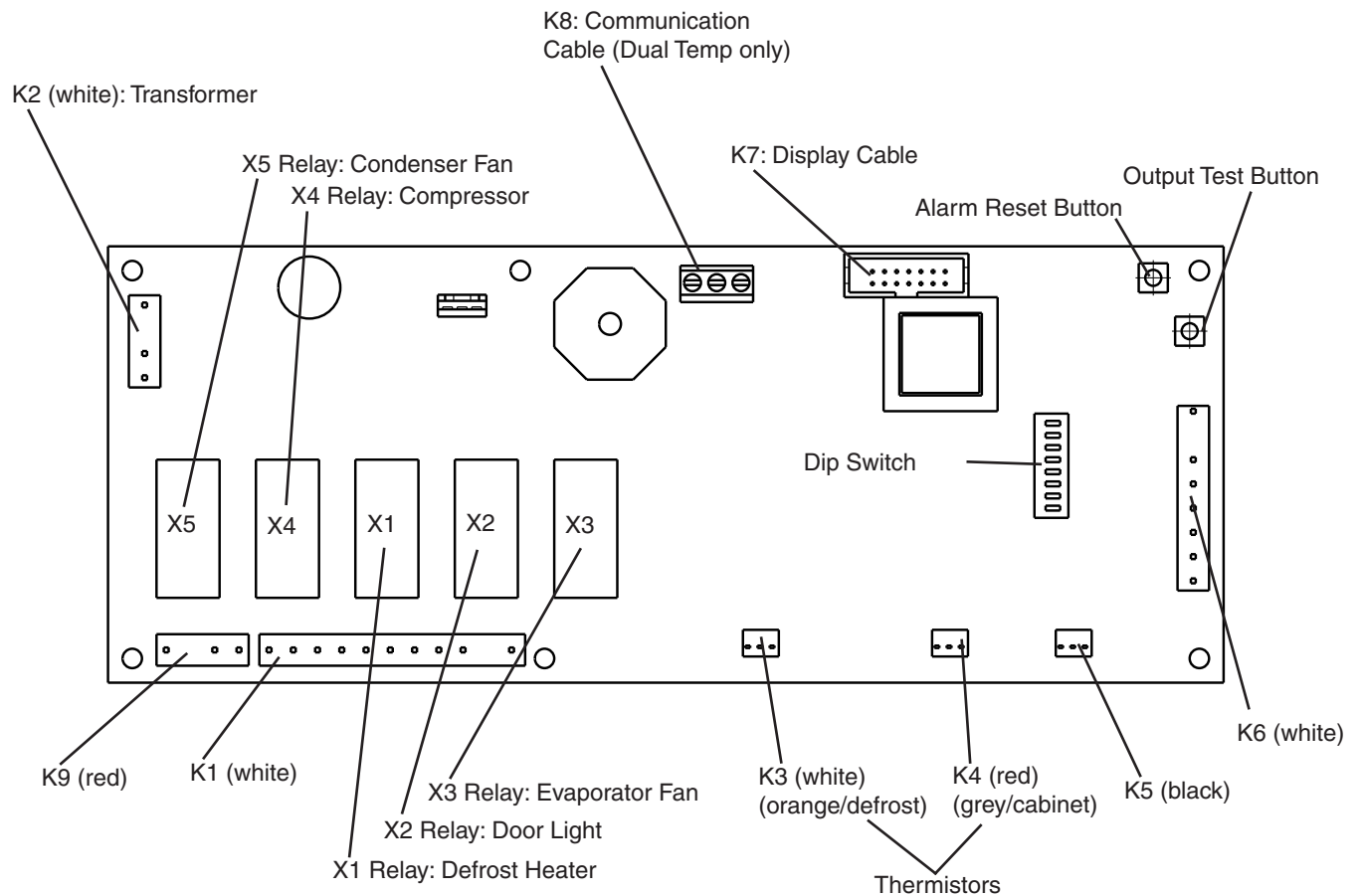
1. Location

The control box, the control panel and an error code label are located behind the front panel. There are two front panel designs. If the front panel swings up when pulled out, swing the panel up and allow both panel hinges to catch securely on both sides of the side panel frame (one on each side). Otherwise, remove the panel by lifting it up approximately one-half inch, then pulling it forward.



2. Control Board Layout

Evercheck Control Board



Note: Each relay has a corresponding light. For the condenser fan (X5), compressor (X4), defrost heater (X1), and door light (X2) relays, the light is "on" when the component is receiving power, and "off" when the component is not receiving power from the board. For the evaporator fan relay (X3), the opposite is the case; the light is "off" when the evaporator fan is receiving power, and "on" when the evaporator fan is not receiving power from the board.

D. Settings and Diagnostics

1. Guarded Access Menu

At start up the control board revision number is displayed. ("r_###", where "###" indicates the current board revision on that model). Three settings can be viewed and changed from this menu; temperature setpoint, defrost frequency, and temperature display scale.

a) *Temperature Setpoint*

The temperature setpoint is the value for the average cabinet temperature. The temperature differential for the compressor to turn on and off is $\pm 3^{\circ}\text{F}$ of the setpoint. For example, setpoint = 36°F , compressor on at 39°F , compressor off at 33°F . If necessary, adjust the setpoint temperature as follows:

- 1) Press and hold the up/down arrows simultaneously for 3 seconds. The current setpoint temperature will be displayed.
- 2) To change the setpoint, press the up/down arrows until the desired value is displayed. For refrigerators the cabinet temperature is adjustable between 36°F and 50°F (2.2°C and 10°C). The factory default is 36°F (2.2°C). For freezers the cabinet temperature is adjustable between -10°F and 25°F (-23.3°C and -3.9°C). The factory default is -3°F (-19.4°C).
- 3) To save the value, press "ENTER" repeatedly until you have cycled through the menu and the unit returns to the normal display mode. If you do not cycle through the menu and no button is pressed in 15 seconds, the display will return to normal and the setpoint will remain unchanged.

b) *Defrost Frequency*

(1a) Defrost for Refrigerators - Control Board Revision 22 and earlier

This unit uses an off-cycle defrost. When a sensor in the evaporator coil reaches 13°F (-10°C) (8°F , -13.3°C for Pass Thru) the unit enters defrost. When the sensor reaches 40°F (4.4°C) the unit ends defrost. No adjustment is available or needed from the control board.

With control board revision 22 and earlier, the cabinet temperature is displayed, even during defrost.

(1b) Defrost for Refrigerators - Control Board Revision 23 and later

This unit uses an off-cycle defrost. When a sensor in the evaporator coil reaches 13°F (-10°C) (8°F , -13.3°C for Pass Thru) the unit enters defrost. When the sensor reaches 40°F (4.4°C) the unit ends defrost. Units with a control board of revision level R23 and later (board revision level can be seen on display at power up) also feature an adjustable time-initiated defrost along with the normal temperature-initiated defrost. The time-initiated defrost is factory set at 0. Before changing this setting, contact the HOSHIZAKI Technical Support Department for recommendations. To change the time-initiated defrost, follow the steps in "(3) Defrost Frequency Adjustment" later in this section. With control board revision 23 and later the cabinet temperature is not displayed during defrost, "dEF" is displayed in its place.

(2) Defrost for Freezer

This unit uses a time-initiated heated defrost. It is preset at the factory to defrost 6 times per day for general conditions. However, if it is determined that this interval does not suit the actual conditions the unit is exposed to, the interval can be changed. To change, see "(3) Defrost Frequency Adjustment" below. Contact the HOSHIZAKI Technical Support Department for recommendations. Please note that since the defrost is a heated defrost, it will have a tendency to raise cabinet temperature. Defrost termination temperature is 100°F (37.8°C).

During freezer defrost, the cabinet temperature is not displayed, "dEF" is displayed in its place.

(3) Defrost Frequency Adjustment

- 1) Press and hold the up/down arrows simultaneously for 3 seconds. Press "ENTER" until the dF frequency is displayed.
- 2) To change the time-initiated defrost frequency, press the up/down arrows until the desired value is displayed. For refrigerators, the defrost frequency is adjustable between 0 and 12 defrosts per 24 hours. For freezers, the defrost frequency is adjustable between 1 and 12 defrosts per 24 hours. The factory default for refrigerators is 0 and the factory default for freezers is 6.
- 3) To save the value, press "ENTER" repeatedly until you have cycled through the menu and the unit returns to the normal display mode. If you do not cycle through the menu and no button is pressed in 15 seconds, the display will return to normal and the defrost frequency will remain unchanged.

Note: The change in the defrost setting will take effect after the next defrost based on the previous setting. If it is desired that this change in interval timing take effect immediately, turn the unit off and back on. The next time-initiated defrost will take effect "x" hours after power is resupplied. For example, if the setting is dF 6, x=4 and the next defrost will take place 4 hours from the time that power is applied.

Primary defrost termination is controlled by the defrost thermistor. However, two additional safeties are also present:

- Time Termination - 1 hour maximum
- Temperature Termination - Measured by a separate device that is in-line with the heaters and independent of the control board

c) Temperature Display Scale (°F or °C)

To change the display scale, follow the steps below.

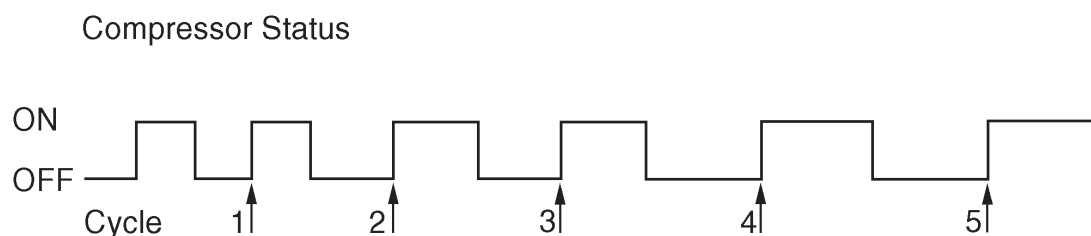
- 1) Press and hold the up/down arrows simultaneously for 3 seconds. Press "ENTER" until "F" or "C" temperature display scale is displayed.
- 2) To change the temperature display, press the up/down arrows until the desired scale is displayed. The factory default is "F".
- 3) Press "ENTER" to save the value and return to normal display mode. If you do not press "ENTER" and no button is pressed in 15 seconds, the display will return to normal and the temperature display scale will remain unchanged.

2. Diagnostics Menu

Access to diagnostic menu is gained by pressing and holding the up/down arrows and the "ENTER" button together for three seconds. Scroll through menu options using the "ENTER" button. Change options using up/down arrows. Reset alarm conditions using the RESET button. To exit, press the "ENTER" button until normal display returns. To exit diagnostic menu at any point in the menu, press and hold the "ENTER" button for three seconds. Diagnostic menu will remain on display for 10 minutes after the last keystroke, then automatically revert to normal operation mode (cabinet temperature display). All information given in degrees will automatically convert to the appropriate programmed scale (°F or °C).

Display	Definition
1 OFF	Displays cabinet temperature.
1 ON	Displays evaporator temperature. Display will automatically revert to cabinet sensor after five minutes.
2 OFF (Freezer Only)	Manual defrost initiation option not activated.
2 ON (Freezer Only)	Manual defrost initiation option activated. Control will begin defrost cycle once user returns to operating mode by cycling through menus. Manual defrost does not eliminate or change times on any scheduled defrost cycles.
3 OFF	Unconditional alarm reset option not activated.
3 ON	Unconditional alarm reset option activated. Changing from 3 OFF to 3 ON resets all alarm display codes simultaneously. This is a complete clearing of alarms.
4 16	Right two digits represent compressor run time in last 24 hours to the nearest hour. Example: 16 hours total run time in the last 24 hours. Value is every 24 hours.
5 50	Right two digits represent compressor on time percentage for the last 5 run cycles. A run cycle begins when the compressor switches from off to on and ends the next time it switches from off to on. (See diagram below). Example: 50% run time over the last 5 run cycles. Value is calculated as follows: Percent on time = On time last 5 cycles/total time last 5 cycles. Value is saved every 5 cycles. Note: "On time last 5 cycles" does not include an on time that was terminated by a defrost cycle or the first on time after a defrost, and "Total time last 5 cycles" does not include an on or off time that was terminated by a defrost cycle, the time spent in defrost or the first on time after a defrost.
6 45	Right two digits represent compressor run time for the last run cycle. Example: 45 minutes of compressor run time in last run cycle.
7 45 (Freezer Only)	Right two digits represent length of time for the last defrost cycle in minutes. Example: 45 minutes in last defrost cycle.
8 67	Right two digits represent highest temperature recorded during the last high temperature alarm. Example: Temperature reached 67 degrees during last high temperature alarm. Display will show "8--" if there has not been a high temperature alarm. Value is saved every 8.5 minutes when in alarm.
9-10	Right three places represent lowest temperature recorded during the last low temperature alarm. Example: Temperature reached -10 degrees during the last low temperature alarm. Display will show "9--" if there has not been a low temperature alarm. Value is saved every 8.5 minutes when in alarm.

The diagram below is an example of run cycle used to calculate run time:



E. Alarm Signals and Countermeasures

If an error occurs, the alarm code and cabinet temperature are displayed in 2-second intervals and an alarm sounds. See the table below for a description of the problem.

Alarm Signals			
Alarm Code	Alarm Sound	Problem	Reset Options
E1	3 beeps every ten seconds	High Temperature Alarm [Freezers with Board Revision Level 10 or Below and All Refrigerators] Cabinet temperature has exceeded set temperature by 10°F (5.6°C) for more than 2 hours. [Freezers with Board Revision Level 11 and Above] Cabinet temperature has exceeded set temperature by 25°F (13.9°C) for more than 4 hours. (When the unit is turned on, the first display reads "r_##", where "##" indicates the board revision level.)	Press "RESET." If temperature has returned to setpoint range, alarm will stop and "E1" will clear. If temperature is not back in range, "RESET" will silence the alarm for 5 minutes. "E1" will continue to flash.
E2	4 beeps every ten seconds	Low Temperature Alarm Cabinet temperature has remained below setpoint by 8°F (4.4°C) for more than 1 hour.	Press "RESET." If temperature has returned to setpoint range, alarm will stop and "E2" will clear. If temperature is not back in range, "RESET" will silence the alarm for 5 minutes. "E2" will continue to flash.
E3 [Freezer Only] (may alternate with "dEF" instead of temperature)	5 beeps every ten seconds	Defrost Alarm Defrost has taken longer than 1 hour. Control board has terminated defrost.	Press "RESET." Alarm will stop and "E3" will clear. If 4 consecutive defrosts take more than 1 hour, a qualified service technician must be called. For the service tech to reset this alarm, press "ALARM RESET" on the control board itself.
E4	6 beeps every ten seconds	High Pressure Alarm Compressor discharge pressure is outside normal operating range. Pressure switch has been triggered 3 or more times in 1 hour. If switch trips 5 times in 1 hour, compressor will be stopped and will not restart.	Press "RESET." If pressure switch resets automatically, alarm will stop and "E4" will clear when pressure switch closes. After 5 pressure trips, the alarm can be silenced for 1 hour by pressing the "RESET". For the service tech to reset this alarm, press "ALARM RESET" on the control board itself.
E6	8 beeps every ten seconds	High Voltage Alarm Line voltage has been too high for at least 10 seconds. To protect the compressor, the compressor has shut down.	Alarm automatically resets when acceptable voltage is detected. Press "RESET" to silence alarm for 5 minutes.
E7	9 beeps every ten seconds	Low Voltage Alarm Line voltage has been too low for at least 10 seconds. To protect the compressor, the compressor has shut down.	Alarm automatically resets when acceptable voltage is detected. Press "RESET" to silence alarm for 5 minutes.

Alarm Signals			
Alarm Code	Alarm Sound	Problem	Reset Options
E8	Constant buzzer	Cabinet Temperature Sensor Malfunction Alarm Cabinet temperature sensor has failed.	After replacing sensor, alarm will reset. During alarm, press "RESET" to silence buzzer for 5 minutes.
E9	Constant buzzer	Defrost Temperature Sensor Malfunction Alarm Defrost temperature sensor has failed.	After replacing sensor, alarm will reset. During alarm, press "RESET" to silence buzzer for 5 minutes.
E10 [Duel Temp Models Only]	10 beeps every ten seconds	Communication Alarm Freezer and refrigerator boards not communicating properly to control compressor delay.	When communication is restored, alarm will reset. During alarm, press "RESET" to silence buzzer for 18 hours.
CF	1 beep every ten seconds	Clogged Filter Alarm Condenser filter needs cleaning.	Clean filter. Allow time for sensor to react, then press "RESET." During alarm, press "RESET" to silence buzzer for 2 hours. If this alarm occurs frequently, discharge temperature is consistently too high. Failure to take action could result in damage to the compressor.
door [Except Glass Door Models]	2 beeps every ten seconds	Display Only: Door open. Both Display and Beeps: Door open longer than 3 minutes.	Close door. During alarm, press "RESET" to silence buzzer for 3 minutes.

Note: Moving the diagnostic menu item 3 to "ON" resets all alarms regardless of whether or not the setpoints are reached. This is a complete clearing of all alarms. See "II.D.2. Diagnostics Menu" for details.

F. Compressor Protector

When a combined temperature/amperage value is above the limit specified by the compressor manufacturer, a protector will operate independently, turning off the compressor. The compressor will restart when this protector has reset.

Note:

1. Compressor protector resets automatically.
2. If the condenser fan is operating and the compressor is off, it is most likely that the protector has operated.

G. Perimeter Frame Heater

This unit is equipped with a perimeter frame heater. This prevents the formation of condensate on the front frame of the unit under high humidity conditions. If operating the unit under conditions where condensate will not form, these heaters may be turned off using the switch on the control box.

H. Glass Door Heater

Glass door units are equipped with a door heater. This prevents the formation of condensate on the glass under high humidity conditions. If operating unit under conditions where condensate will not form, these heaters may be turned off using the switch on the control panel.

III. Service Diagnosis

A. Diagnosis Chart

Problem	Possible Cause		Remedy
[1] Compressor will not start—no current draw.	a) Power Supply	1. "OFF" position.	1. Move to "ON" position.
		2. Loose connection.	2. Tighten.
		3. Failure.	3. Call electrician.
		4. 208/230 volt; wrong leg to power control board.	4. Correct wiring.
	b) Cord and Plug	1. Defective.	1. Replace.
	c) Ground Fault Circuit Interrupter	1. Tripped.	1. Check and reset.
		2. Defective.	2. Replace.
	d) Transformer	1. Open coil winding.	1. Check continuity and replace.
	e) Wiring to Control Board	1. Loose connection.	1. Tighten.
		2. Faulty.	2. Check continuity and replace.
	f) Voltage	1. Too low.	1. Call electrician.
		2. Too high.	2. Call electrician.
		3. 208/230 volt; wrong leg to power control board.	3. Correct wiring.
	g) High Pressure Switch	1. Bad contacts.	1. Check continuity and replace.
		2. Has operated 5 or more times in one hour. (E4 should be displayed)	2. Press "RESET" on control board. Determine why pressure switch is activated.
h) Thermistor	1. Defective.	1. Replace.	
i) Control Board	1. Defective.	1. Replace control board.	
j) Compressor Overload	1. Defective. (Contacts open)	1. Replace.	
k) Compressor	1. Open windings.	1. Check continuity and replace.	
l) Relay (Condensing Unit)	1. Bad contacts.	1. Replace.	
	2. Open coil winding.	2. Check and replace.	
[2] Compressor will not run—draws current and trips on overload.	a) Voltage	1. Too low.	1. Call electrician.
	b) Start Relay	1. Bad contacts.	1. Replace.
		2. Open coil windings.	2. Replace.
	c) Compressor	1. Locked rotor.	1. Replace.
d) Start Capacitor	1. Defective.	1. Check and replace.	
[3] Compressor runs intermittently and trips on overload.	a) Voltage	1. Too low.	1. Call electrician.
		2. Too high.	2. Call electrician.
	b) Condenser Filter	1. Clogged.	1. Clean filter.
	c) Refrigerant Line or Component	1. Plugged or restricted.	1. Clean and replace drier.
	d) Condenser Fan Motor	1. Failed	1. Replace.
	e) Refrigerant	1. Overcharged.	1. Evacuate and recharge.
		2. Non-condensibles in system.	2. Evacuate and recharge.
	f) Location of Unit	1. Restricted air flow to condenser.	1. Move unit or increase ventilation.
	g) Boards not Communicating (Dual Temp units only)	1. Communication disconnected.	1. Restore communication.
2. Wrong dip switch setting; #6 off, either board.		2. Turn #6 on.	

Problem	Possible Cause		Remedy
[4] Cabinet inside temperature too high; compressor will not start.	a) Thermistor	1. Defective.	1. Check and replace.
	b) Compressor Relay	1. Defective. (LED 4 is ON. 115V to relay coil.)	1. Check and replace.
	c) Circuit Board	1. Defective. (LED 4 is OFF. Compressor relay coil not energized at 115V.)	1. Replace.
[5] Cabinet temperature too high.	a) Setpoint	1. Incorrect.	1. Correct setpoint. Factory defaults: Freezer -3°F Refrigerator 36°F
	b) Door	1. Not sealing, or open for long intervals.	1. Check for sealing, check for door open at time of warm cabinet temperature.
	c) Defrost	1. Not enough defrosts occurring per day. Operation in humid conditions.	1. Freezer: See "D.1.b) Defrost Frequency." 2. Refrigerator: See "D.1.b) Defrost Frequency." 3. Check thermistor reading and position.
	d) Refrigerant	1. Leak.	1. Repair leak and recharge.
	e) Fan Motor	1. Defective.	1. Check and replace.
	f) Air Filter	1. Clogged.	1. Clean.
	g) Condenser	1. Dirty.	1. Clean.
	h) Thermistor	1. Defective.	1. Check and replace.
	i) Control Board	1. Defective.	1. Replace.
[6] Cabinet temperature display indicator does not illuminate properly.	a) Board Connection	1. Bad connection between control board and display board.	1. Check connection.
	b) Display Board	1. Defective.	1. Replace.
	c) Control Board	1. Defective.	1. Replace.
[7] Cabinet inside temperature too low.	a) Thermistor	1. Defective.	1. Check and replace.
	b) Compressor Relay	1. Defective; contacts welded.	1. Replace.
	c) Circuit Board	1. Defective.	1. Replace.
[8] Evaporator does not defrost completely.	a) Defrost Thermistor	1. Defective.	1. Check and replace.
	b) Defrost	1. Not enough defrosts occurring per day. Operation in humid conditions.	1. Freezer: See "D.1.b) Defrost Frequency." 2. Refrigerator: See "D.1.b) Defrost Frequency." 3. Check thermistor reading and position.
	c) Defrost Heaters	1. Defective. (E3 should be displayed)	1. Freezer: Replace heaters. Reset E3 by pressing control board "ALARM RESET."
	d) Safety Defrost Thermostat	1. Defective, turning off heaters prematurely, or fused open. (E3 should be displayed)	1. Freezer: Replace safety defrost thermostat. Reset E3 by pressing control board "ALARM RESET."

Problem	Possible Cause		Remedy
[9] Defrost cycle lasts too long.	a) Defrost Thermistor	1. Defective.	1. Replace.
	b) Control Board	1. Defective.	1. Replace.
	c) Defrost	1. Not enough defrosts occurring per day. Operation in humid conditions.	1. Freezer: See "D.1.b) Defrost Frequency." 2. Refrigerator: See "D.1.b) Defrost Frequency." 3. Check thermistor reading and position.
	d) Defrost Heaters	1. Defective. (E3 should be displayed)	1. Freezer: Replace heaters. Reset E3 by pressing control board "ALARM RESET."
	e) Safety Defrost Thermostat	1. Defective, turning off heaters prematurely, or fused open. (E3 should be displayed)	1. Freezer: Replace safety defrost thermostat. Reset E3 by pressing control board "ALARM RESET."
[10] Condensate water overflow.	a) Cabinet Contents	1. Loading of large volumes of warm, moist, uncovered product.	1. Cover product with plastic wrap.
	b) Location of Unit	1. Unit located near high humidity source such as fryer, steamer, etc.	1. Relocate.
	c) Seals	1. Poor sealing around evaporator, door gaskets.	1. Adjust or replace.
	d) Environment	1. Extreme environment and door-opening conditions.	1. Adjust conditions.
[11] Abnormal noise	a) Fasteners	1. Loose fasteners allow vibration of part.	1. Tighten fasteners.
	b) Compressor	1. Problem with mount.	1. Properly mount compressor. Replace any missing grommets.
		2. Floodback to compressor.	2. Check for signs of floodback to compressor. Evacuate and recharge if necessary.
		3. Defective.	3. Replace.
	c) Fan	1. Fan blade loose.	1. Adjust and tighten.
		2. Defective motor.	2. Replace.
	d) Relay	1. Chattering.	1. Replace.

B. Thermistor Check

Thermistors (semiconductors) are used for the cabinet control sensor and defrost termination sensor. The resistance varies depending on temperature. No adjustment is required. If necessary, check for resistance between thermistor leads and visually check the thermistor mounting.

Temperature		Resistance kΩ
°F	°C	
0	-17.8	13.69
10	-12.2	10.48
32	0.0	6.00
50	10.0	3.96
70	21.1	2.57
90	32.2	1.71

To check a thermistor for resistance, follow the steps below.

- 1) Turn power off and disconnect connectors K3 and K4 on the board.
- 2) Remove the thermistor.
- 3) Immerse the thermistor sensor portion in a glass containing ice and water for 2 or 3 minutes.
- 4) Check for a resistance between thermistor leads. Normal reading is within 3.5 to 7 kΩ. Replace the thermistor if it exceeds the normal reading.

IV. Removal and Replacement of Components

IMPORTANT

1. Ensure all components, fasteners and thumbscrews are securely in place after the equipment is serviced.
2. The Polyol Ester (POE) oils used in R-404A units can absorb moisture quickly. Therefore it is important to prevent moisture from entering the system when replacing or servicing parts.
3. Always install a new drier every time the sealed refrigeration system is opened. Do not replace the drier until after all other repair or replacement has been made.
4. Do not leave the system open for longer than 15 minutes when replacing or servicing parts.

A. Service for Refrigerant Lines

WARNING

Do not use R-404A as a mixture with pressurized air for leak testing. Refrigerant leaks can be detected by charging the unit with a trace of refrigerant, raising the pressure with nitrogen and using an electronic leak detector.

1. Refrigerant Recovery

The unit is provided with refrigerant access valves. Using proper refrigerant practices recover the refrigerant from the access valves and store it in an approved container. Do not discharge the refrigerant into the atmosphere.

2. Brazing

WARNING

1. Refrigerant R-404A itself is not flammable at atmospheric pressure and temperatures up to 176°F (80°C).
2. Refrigerant R-404A itself is not explosive or poisonous. However, when exposed to high temperatures (open flames) R-404A can be decomposed to form hydrofluoric acid and carbonyl fluoride both of which are hazardous.
3. Always recover the refrigerant and store it in an approved container. Do not discharge the refrigerant into the atmosphere.
4. Do not use silver alloy or copper alloy containing arsenic.
5. Do not use R-404A as a mixture with pressurized air for leak testing. Refrigerant leaks can be detected by charging the unit with a trace of refrigerant, raising the pressure with nitrogen and using an electronic leak detector.

- 1) Always install a new drier every time the sealed refrigeration system is opened. Do not replace the drier until after all other repair or replacement has been made. Install the new drier with the arrow on the drier in the direction of the refrigerant flow.
- 2) Braze all fittings while purging with nitrogen gas flowing at a pressure of 3 to 4 PSIG.
- 3) Check for leaks using nitrogen gas (140 PSIG) and soap bubbles. Do not use R-404A as a mixture with pressurized air for leak testing.

Note: Because the pipes in the evaporator case are specially coated to resist corrosion, it is important to make connections outside the evaporator case when possible. If it is necessary to braze inside the evaporator case, use sandpaper to remove the coating from the brazing connections before unbrazing the components.

3. Evacuation and Recharge

- 1) Attach a vacuum pump to the system. Be sure the charging hoses are connected to both high and low-side access valves.

IMPORTANT

The vacuum level and vacuum pump may be the same as those for current refrigerants. However, the rubber hose and gauge manifold to be used for evacuation and refrigerant charge should be exclusively for POE oils.

- 2) Turn on the vacuum pump. Never allow the oil in the vacuum pump to flow backward.
- 3) Allow the vacuum pump to pull down to a 29.9" Hg vacuum. Evacuating period depends on pump capacity.
- 4) Close the low-side valve and high-side valve on the service manifold.
- 5) Disconnect the vacuum pump, and attach a refrigerant service cylinder to the high-side line. Remember to loosen the connection, and purge the air from the hose. See the nameplate for the required refrigerant charge. Hoshizaki recommends only virgin refrigerant or reclaimed refrigerant which meets ARI Standard No. 700-88 be used.
- 6) A liquid charge is recommended for charging an R-404A system. Invert the service cylinder and place it on scales. Open the high-side, service manifold valve.
- 7) Allow the system to charge with liquid until the proper charge weight is met.
- 8) If necessary, add any remaining charge to the system through the low-side. Use a throttling valve or liquid dispensing device to add the remaining liquid charge through the low-side access port with the unit running.
- 9) Close the two refrigerant access valves, and disconnect the hoses and service manifold.
- 10) Cap the access valves to prevent a possible leak.

B. Removal and Replacement of Compressor

IMPORTANT

Always install a new drier every time the sealed refrigeration system is opened. Do not replace the drier until after all other repair or replacement has been made. Install the new drier with the arrow on the drier in the direction of the refrigerant flow.

Note: When replacing a compressor with a defective winding, be sure to install the new start capacitor and start relay supplied with the replacement compressor. Due to the ability of the POE oil in the compressor to absorb moisture quickly, the compressor must not be opened more than 15 minutes for replacement or service. Do not mix lubricants of different compressors even if both are charged with R-404A, except when they use the same lubricant.

- 1) Unplug the unit from the electrical outlet.
- 2) Remove the panels.
- 3) Recover the refrigerant and store it in an approved container.
- 4) Remove the terminal cover on the compressor, and disconnect the compressor wiring.
- 5) Remove the hold-down bolts, washers, rubber grommets and sleeves.
- 6) Remove the discharge and suction pipes.
- 7) Remove the compressor. Unpack the new compressor package.
- 8) Attach the rubber grommets of the prior compressor.
- 9) Place the compressor in position and secure it using the bolts and washers
- 10) Remove the drier, then place the new drier in position.
- 11) Remove plugs from the suction, discharge and process pipes.
- 12) Braze all fittings while purging with nitrogen gas flowing at a pressure of 3 to 4 PSIG.
- 13) Check for leaks using nitrogen gas (140 PSIG) and soap bubbles.
- 14) Evacuate the system, and charge it with refrigerant. See the nameplate for the required refrigerant charge.
- 15) Connect the terminals and replace the terminal cover in its correct position.
- 16) Replace the panels in their correct positions.
- 17) Plug the unit back in.

C. Removal and Replacement of Expansion Valve

IMPORTANT

Sometimes moisture in the refrigeration circuit exceeds the drier capacity and freezes up at the expansion valve. Always install a new drier every time the sealed refrigeration system is opened. Do not replace the drier until after all other repair or replacement has been made. Install the new drier with the arrow on the drier in the direction of the refrigerant flow.

- 1) Unplug the unit from the electrical outlet.
- 2) Remove the panels.
- 3) Recover the refrigerant and store it in an approved container.
- 4) Remove the insulation and the expansion valve bulb on the suction line.
- 5) Remove the expansion valve cover and disconnect the expansion valve. Place the new expansion valve in position.
- 6) Remove the drier, then place the new drier in position.
- 7) Braze all fittings while purging with nitrogen gas flowing at a pressure of 3 to 4 PSIG.

WARNING

Always protect the valve body by using a damp cloth to prevent the valve from overheating. Do not braze with the valve body exceeding 250°F (121°C).

- 8) Check for leaks using nitrogen gas (140 PSIG) and soap bubbles.
- 9) Evacuate the system, and charge it with refrigerant. See the nameplate for the required refrigerant charge.
- 10) Attach the expansion valve bulb to the suction line in the same location as the previous bulb. The bulb should be at the 12 o'clock position on the tube. Be sure to secure the bulb with the clamp and holder and to insulate it.
- 11) Place the expansion valve cover in position.
- 12) Replace the panels in their correct positions.
- 13) Plug the unit back in.

D. Removal and Replacement of Evaporator

IMPORTANT

Always install a new drier every time the sealed refrigeration system is opened. Do not replace the drier until after all other repair or replacement has been made. Install the new drier with the arrow on the drier in the direction of the refrigerant flow.

- 1) Unplug the unit from the electrical outlet.
- 2) Remove the panels and the top cover over the evaporator.
- 3) Recover the refrigerant and store it in an approved container.
- 4) Remove the insulation tubing, and disconnect the evaporator inlet and outlet tubing. Elevate evaporator to avoid overheating of evaporator housing.
- 5) Place the new evaporator in position.
- 6) Remove the drier, then place the new drier in position.
- 7) Check for leaks using nitrogen gas (140 PSIG) and soap bubbles.
- 8) Evacuate the system, and charge it with refrigerant. See the nameplate for the required refrigerant charge.
- 9) Replace the removed parts in the reverse order of which they were removed.
- 10) Replace the top cover and the panels in their correct positions.
- 11) Plug the unit back in.

E. Removal and Replacement of Evaporator Fan Shroud Assembly

1. Note for RH1-SSB(-HD), RH1-SSB-GD and Refrigerator Side of RFH2-SSB(-HD)

For RH1-SSB(-HD) aux. code M5____B and earlier, RH1-SSB-GD aux. code L0 and earlier, and refrigerator side of RFH2-SSB(-HD) aux. code M5 and earlier, there was a non-interchangeable part change. If your unit falls into this time frame and you need to replace parts in the evaporator fan shroud assembly, review the scenarios below and follow the appropriate solution.

- a. Unit's evaporator shroud has only two holes or has four holes but only the lower two holes are threaded.

Solution: Order HS-3541 that contains a motor (Shinko), brackets, blade, and all fasteners required for mounting to the shroud using the existing threaded holes. For details, see the unit's parts list.

- b. Unit's evaporator shroud has four threaded holes and a Jakel motor.

Solution: Order a new motor (Shinko), brackets, fan blade, and the appropriate fasteners listed in the unit's parts list. Attach the brackets to the upper two holes on the shroud.

2. Note for FH1-SSB(-HD) and Freezer Side of RFH2-SSB(-HD) and RFH3-SSB(-HD)

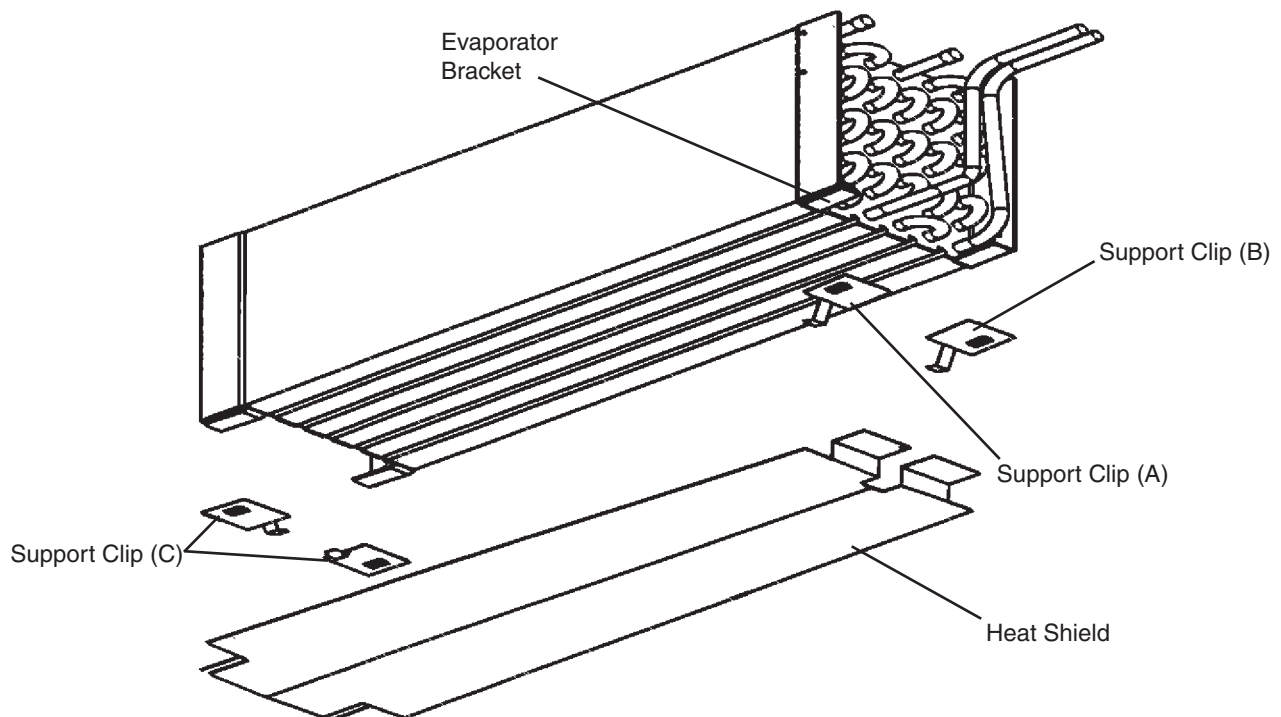
For FH1-SSB(-HD) and freezer side of RFH2-SSB(-HD) and RFH3-SSB(-HD) aux. code M5 and earlier, there was a non-interchangeable part change. If your unit falls into this time frame and you need to replace parts in the evaporator fan shroud assembly, order HS-3545 and replace the entire assembly.

F. Removal and Replacement of Heat Shield on FH2-SSB(-HD)

The heat shield was added to the evaporator assembly at auxiliary code M6. To replace the shield, follow the instructions below.

- 1) Remove the original heat shield by lifting the evaporator up and sliding the heat shield out.
- 2) Install the new heat shield in the rotocast drain pan.
- 3) Slide the clips onto the end plate on the evaporator. Install the two Support Clip (C) clips onto the hairpin end of the evaporator as shown in the illustration.
- 4) Install Support Clip (A) and Support Clip (B) onto the return bend end (same side as inlet and outlet tubes) as shown in the illustration.

Note: All clips are shown in the illustration in the correct orientation for installation. Slide the parts onto the evaporator brackets using the tabs on the support clips.



G. Removal and Replacement of Door Gasket

Remove old gasket by pulling it directly out of the vinyl gasket retainer. Thoroughly clean the gasket area with mild soap and water prior to installing the new gasket. The new gasket should be installed by assembling it at the corners first, then working toward the center at the top, bottom and sides. The arrow-shaped portion of the gasket should be firmly seated in the retainer groove for proper assembly. This can be checked by lifting the edge of the gasket and observing the engagement.

CAUTION

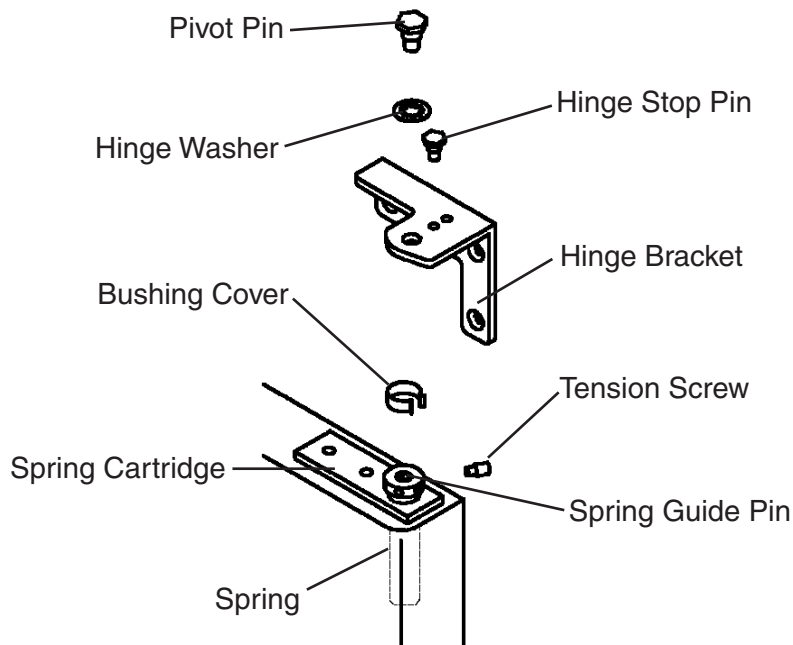
In order to get a proper gasket fit, it is important not to stretch gasket material during assembly.

H. Removal and Replacement of Door Closure Spring

CAUTION

Wear eye protection and use caution when removing the tension screw (step 3).

- 1) Open the front panel assembly to gain access to the upper hinge brackets.
- 2) Remove the bushing cover from the top of the spring guide pin.



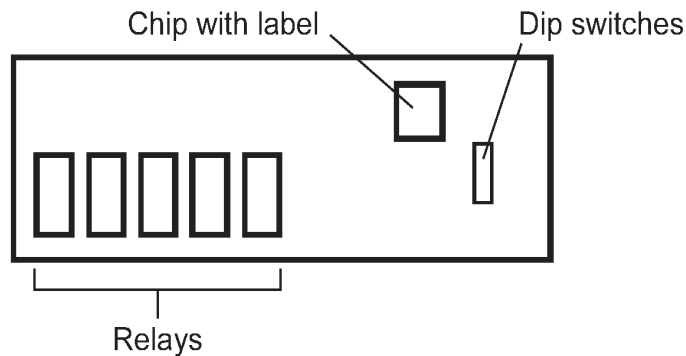
- 3) Insert a small drift pin or long leg of an allen wrench into one of the threaded tension screw holes in the spring guide pin. Turn the spring guide pin to access the tension screw. Securely hold the spring guide pin in this position while removing the tension screw. Carefully rotate (walk) the spring guide pin to release spring tension.
- 4) Use a 1/2" socket wrench to remove the top pivot pin/hinge washer from the top hinge bracket.
- 5) Firmly grasp the door and pull it forward at the top. Then raise the door, disengaging it from the bottom pivot pin.
- 6) Remove the spring guide pin and spring from the spring cartridge.
- 7) Replace the spring with one of the same color. Be certain that the spring ends are engaged in both the spring cartridge and the top of spring guide pin.
- 8) Repeat steps 1 through 7 for additional doors as necessary.
- 9) Reverse procedure to reassemble door(s) to cabinet.

Side	Color	Part Number
Left	Blue	4A3312-02
Right	Black	4A3312-01

Note: Tighten spring guide pin one complete rotation to set proper spring tension.

I. Removal and Replacement of Control Board

- 1) Unplug the unit from the electrical outlet.
- 2) Remove the front panel.
- 3) Remove the control box cover.
- 4) Unhook all wiring connections to the control board.
- 5) Remove the control board by carefully squeezing the tips of the board mounts.
- 6) Identify the board revision of the new control board. This will be marked on the micro-controller chip label as shown in the diagram below.

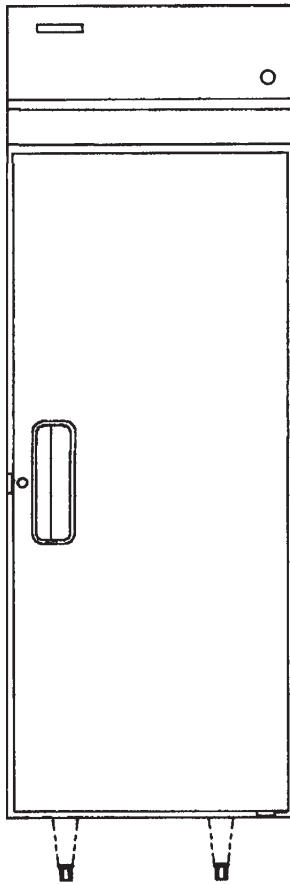


- 7) Rev. 11 or lower control boards will work only with the round light switches (4A0487-01). These switches are on front of the cabinet behind the doors.
- 8) Rev. 12 and higher will work on both round and square door switches. For round switches, dip switch #3 must be flipped on. For square switches, leave the dip switch off.
- 9) Verify dip switches are set correctly. See the unit's wiring label for settings.
- 10) Replace the board in the control box taking care not to damage it.
- 11) Hook all wiring connections back to the control board.
- 12) Replace all panels.
- 13) Plug the unit back in.

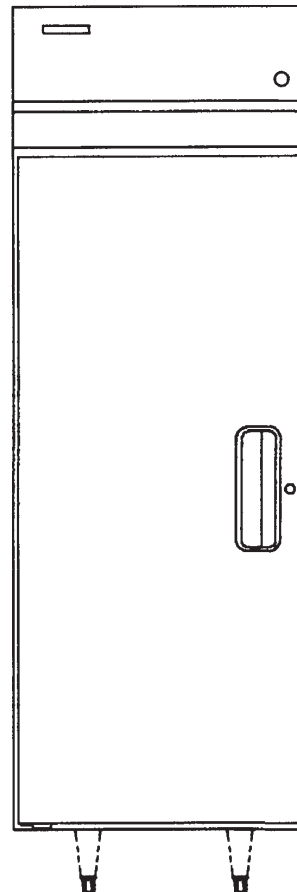
J. Door Re-Hinging (except glass doors)

The door on any unit is reversible as shown below. The instructions for re-hinging are provided in the kits listed below.

Convert	Door Type	Kit Number
Right to Left	Full	HS-3527
Left to Right	Full	HS-3528
Right to Left	Half	HS-3529
Left to Right	Half	HS-3530



The door hinges on the right cabinet wall.



The door is rotated and hinged on the left cabinet wall.

V. Cleaning Instructions

1. Stainless Steel Exterior

Wipe the exterior occasionally with a clean, soft cloth. Use a damp cloth containing a neutral cleaner to wipe off oil or dirt build up.

2. Stainless Steel Interior

Spills should be wiped up promptly to avoid unpleasant odors. The cabinet interior should be cleaned periodically with a mild soap or detergent.

3. Door Gaskets

Door gaskets should be cleaned regularly with mild soap and water to remove dirt and grease.

4. Air Filter

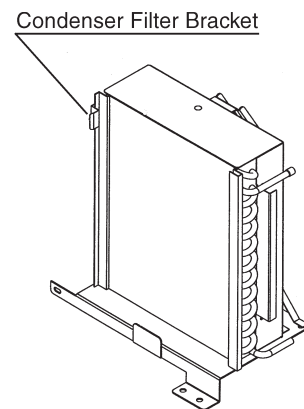
Check the filter at least once a month. If it is dirty, use warm water and a neutral cleaner to wash the filter (See "II.E.10. Clogged Filter Alarm"). The filter is located behind the upper front panel. This filter can be removed by bending the condenser filter bracket back, sliding the filter up and out and allowing the bracket spring back into place.

5. Condenser

Check the condenser once a year and use a brush or vacuum cleaner to clean the unit as required.

6. Glass Door

Wipe the exterior occasionally with a clean, soft cloth. Use a damp cloth containing a neutral cleaner to wipe off oil or dirt build up.

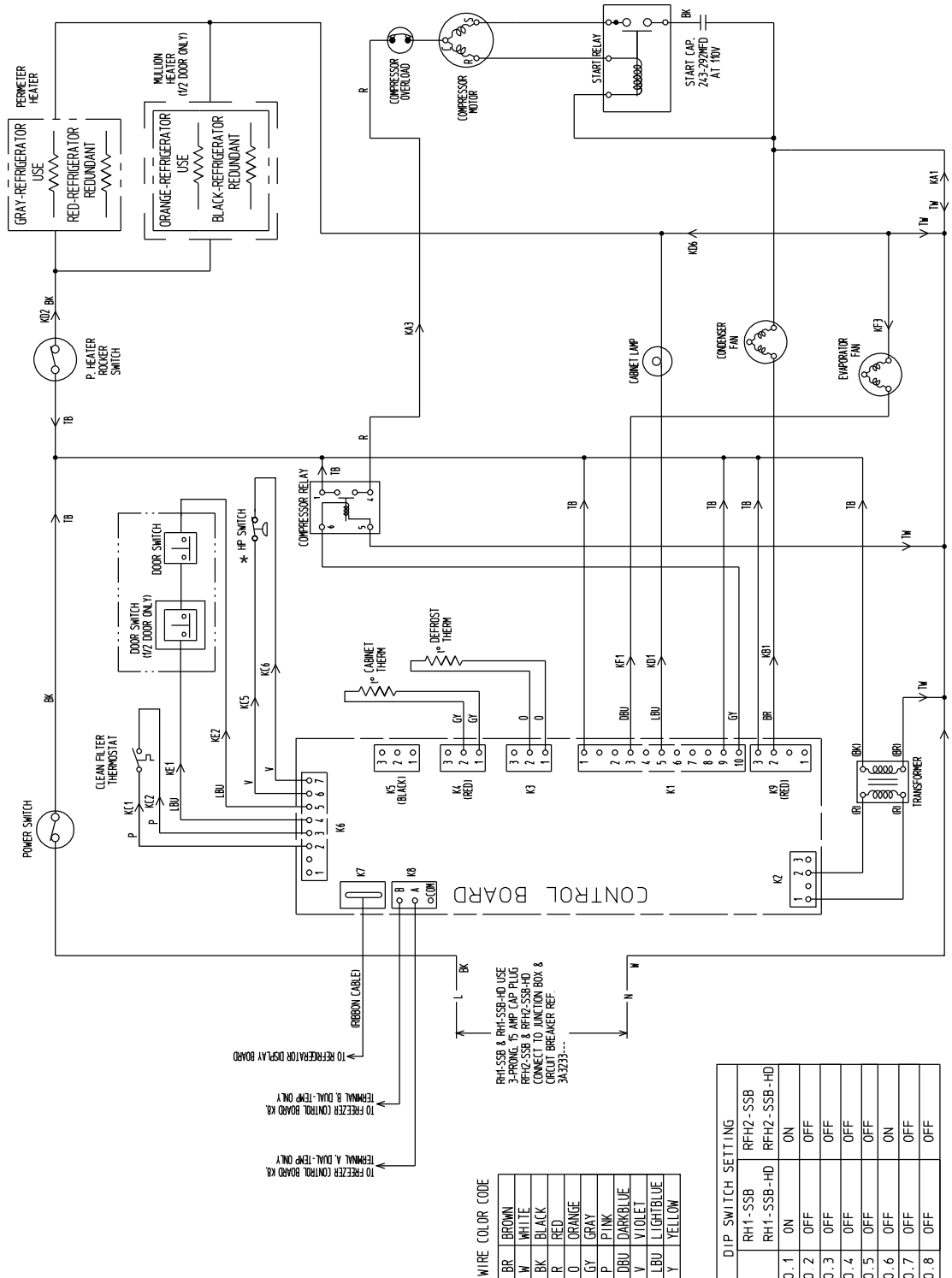


A1. RH1-SSB(-HD) (auxiliary code L5 and earlier)



Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

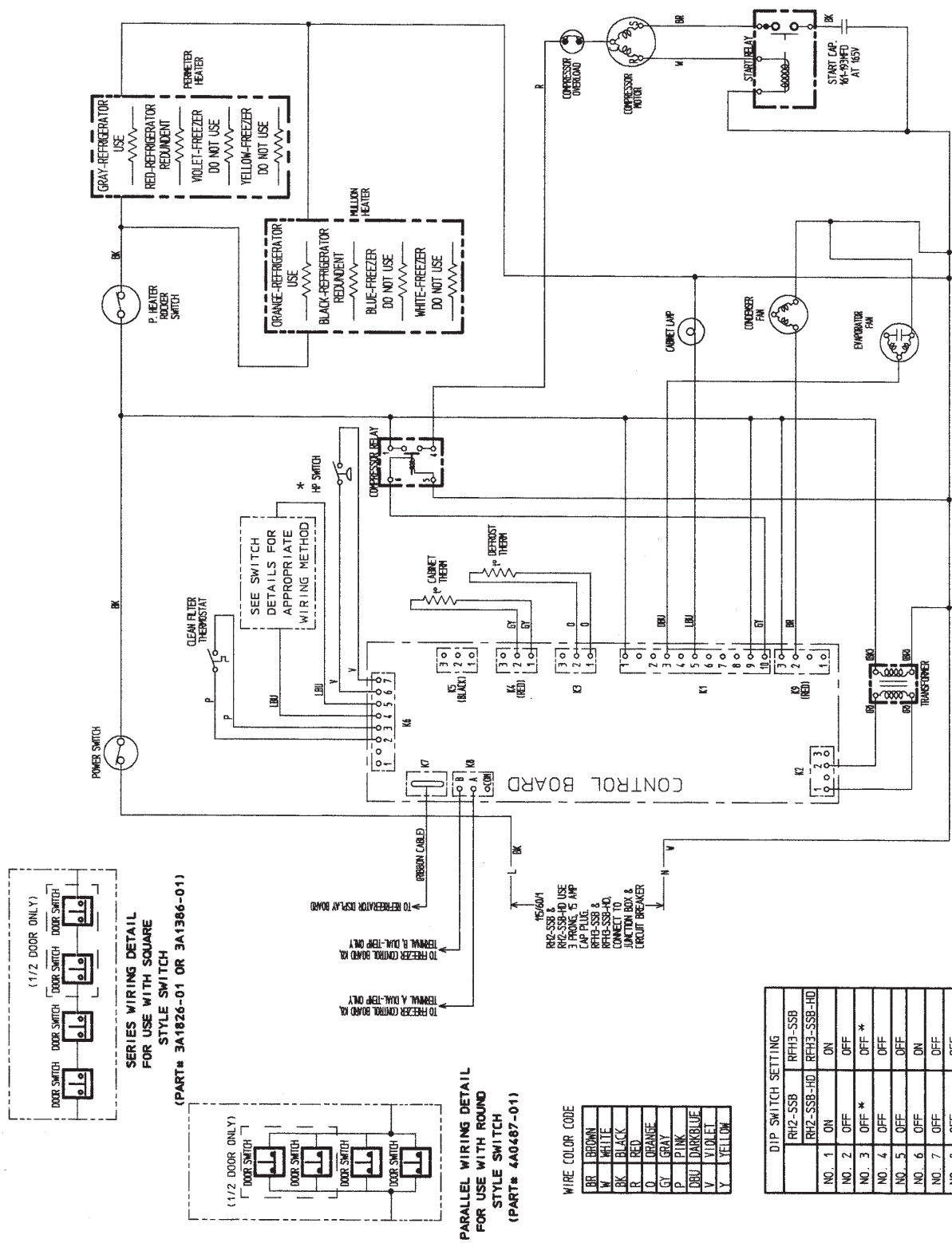
A2. RH1-SSB(-HD) (auxiliary code M5 and later)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

B1. RH2-SSB(-HD) (auxiliary code L5 and earlier)



* Pressure Switch (normally open)		
Cut-out (closed)	450±10 PSIG	
Cut-in (open)	320±10 PSIG	

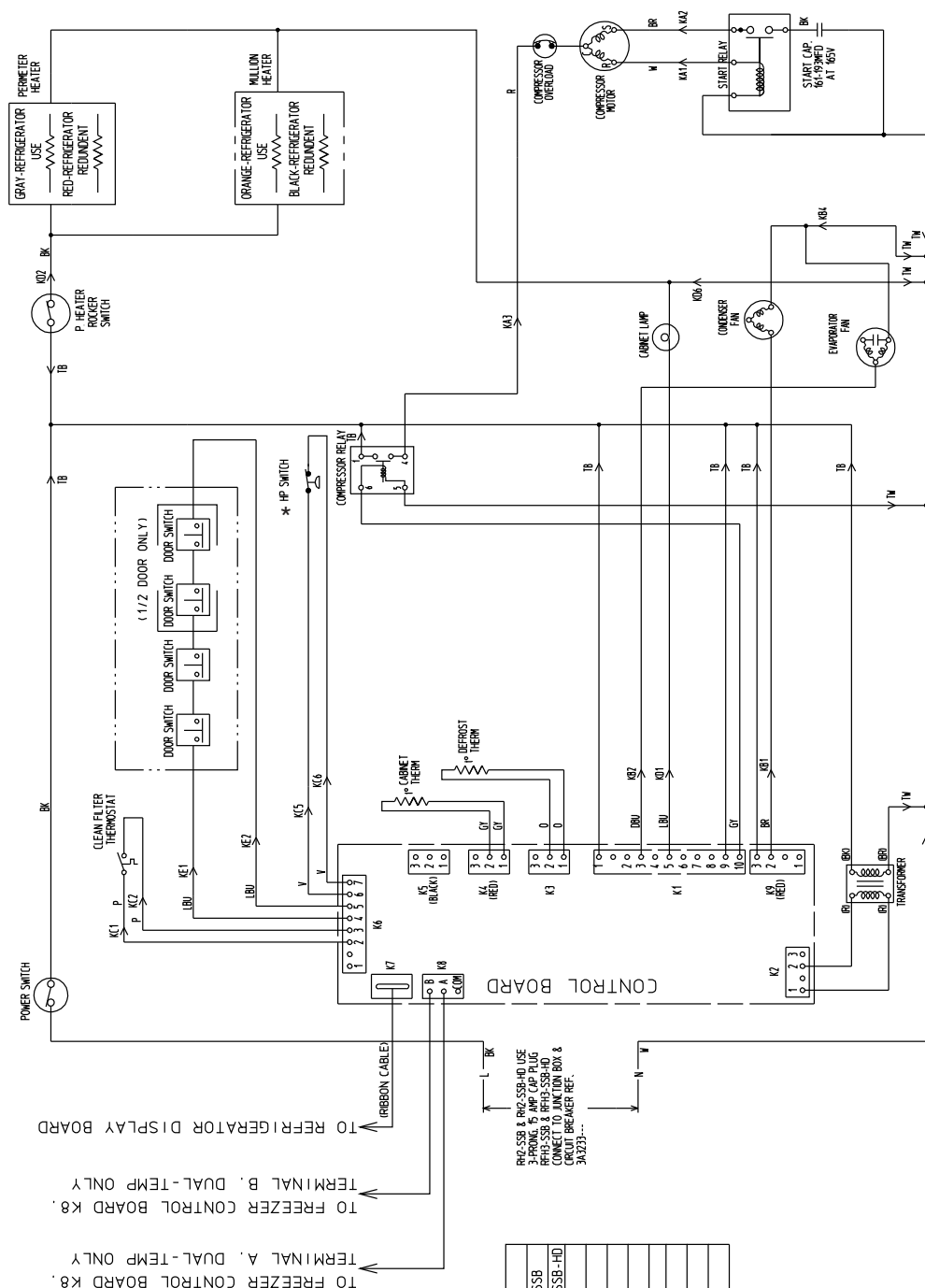
DIP SWITCH SETTING	
RH2-SSB	RH3-SSB
RH2-SSB-HD	RH3-SSB-HD
NO. 1	ON
NO. 2	OFF
NO. 3	OFF *
NO. 4	OFF
NO. 5	OFF
NO. 6	OFF
NO. 7	OFF
NO. 8	OFF

* FOR USE OF PART# 3A1386-01 OR 3A1826-01 SQUARE LIGHTSWITCH DIP SWITCH MUST BE SET TO OFF. FOR USE OF PART# 4A0487-01 ROUND LIGHTSWITCH DIP SWITCH MUST BE SET TO ON.

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

B2. RH2-SSB(-HD) (auxiliary code M5 and later)



WIRE COLOR CODE

BR	BROWN
W	WHITE
BK	BLACK
R	RED
O	ORANGE
GY	GRAY
P	PINK
DBU	DARKBLUE
V	VIOLET
Y	YELLOW

DIP SWITCH SETTING

	RH2-SSB	RH3-SSB
NO. 1	ON	ON
NO. 2	OFF	OFF
NO. 3	OFF	OFF
NO. 4	OFF	OFF
NO. 5	OFF	OFF
NO. 6	OFF	ON
NO. 7	OFF	OFF
NO. 8	OFF	OFF

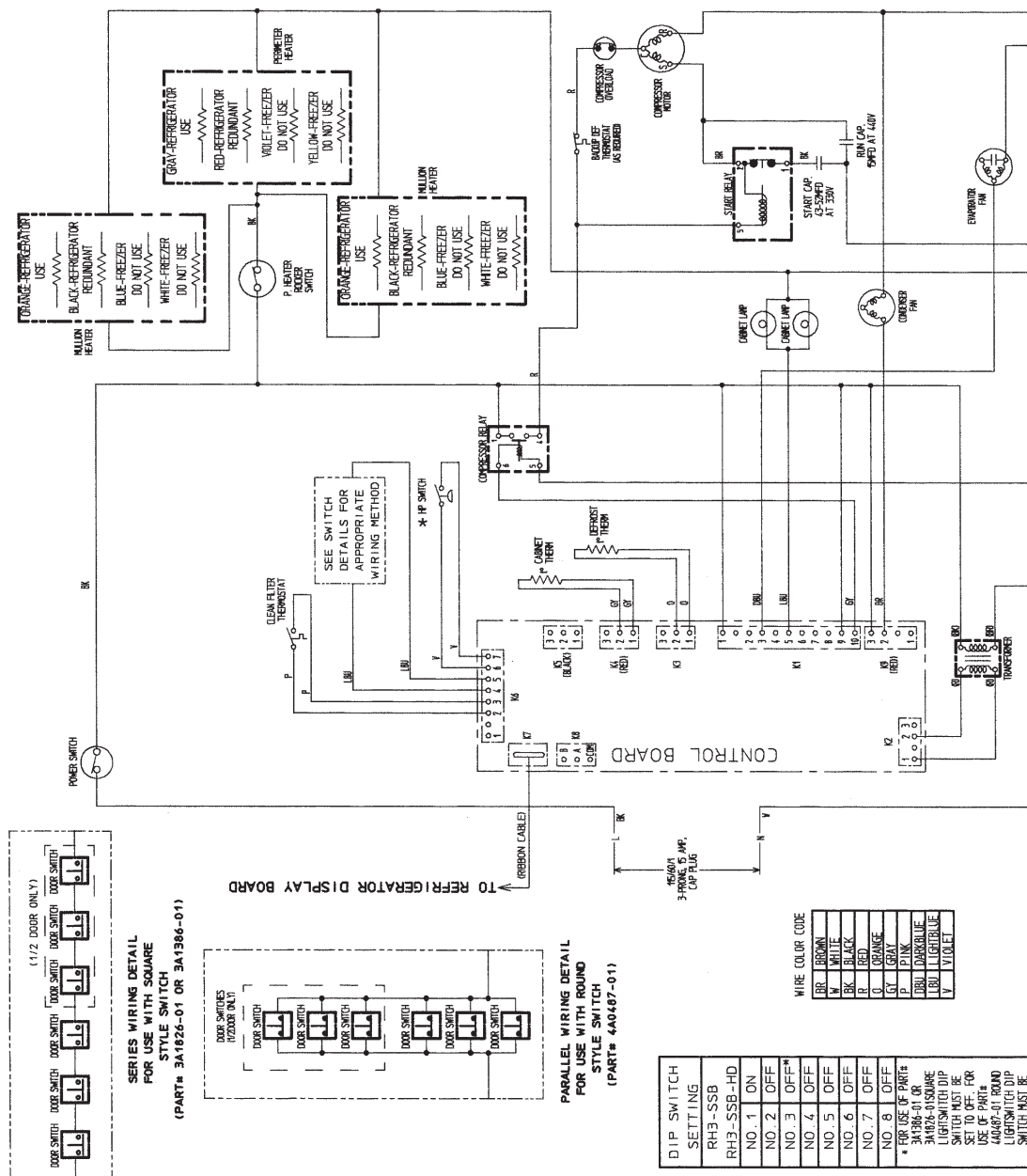
* Pressure Switch (normally closed)

Cut-out (open)	490±10 PSIG
Cut-in (closed)	370±20 PSIG

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

C1. RH3-SSB(-HD) (auxiliary code L5 and earlier)

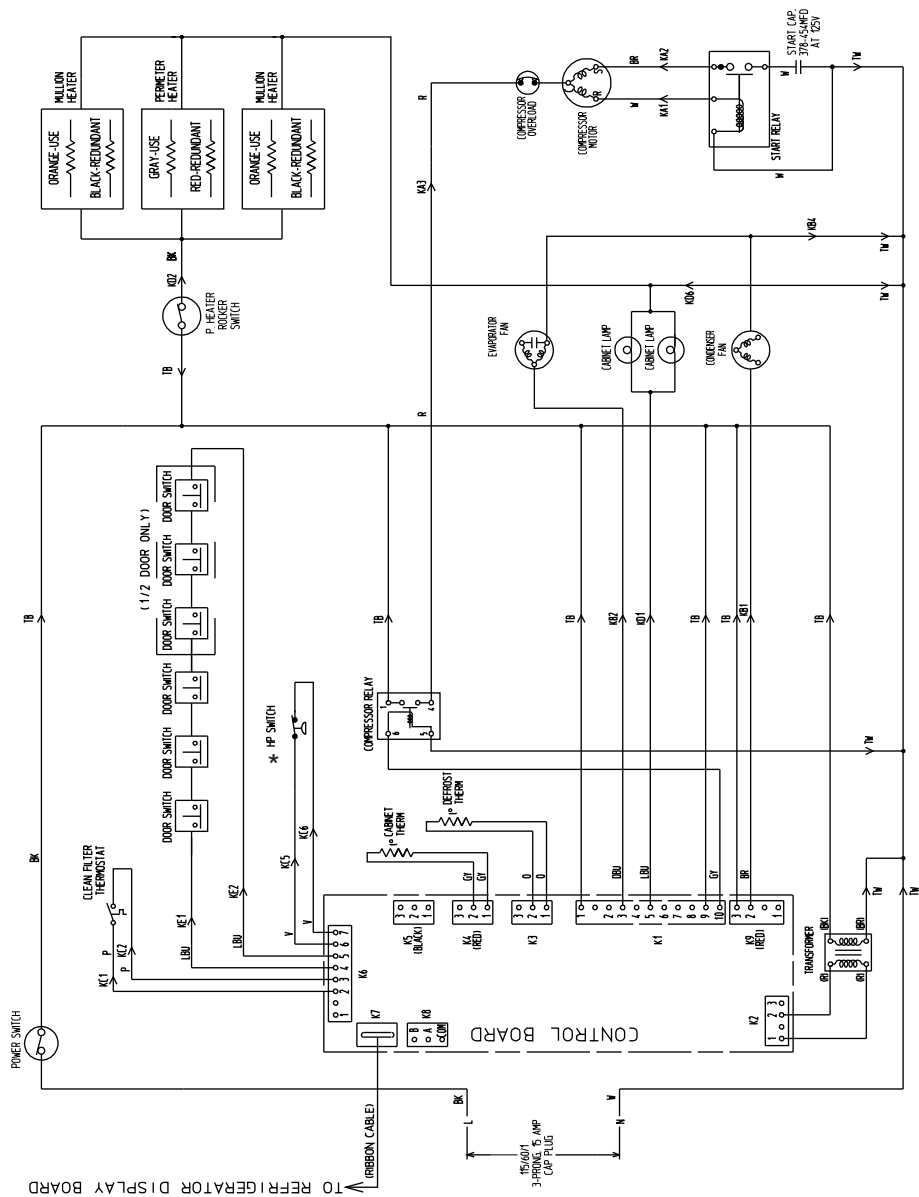


* Pressure Switch (normally open)		
	Aux code H0, J0	Aux code J1 to L5
Cut-out (closed)	450±10 PSIG	500±10 PSIG
Cut-in (open)	320±10 PSIG	330±10 PSIG

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

C2. RH3-SSB(-HD) (auxiliary code M5 and later)



* Pressure Switch (normally closed)		
Cut-out (open)	490±10 PSIG	
Cut-in (closed)	370±20 PSIG	

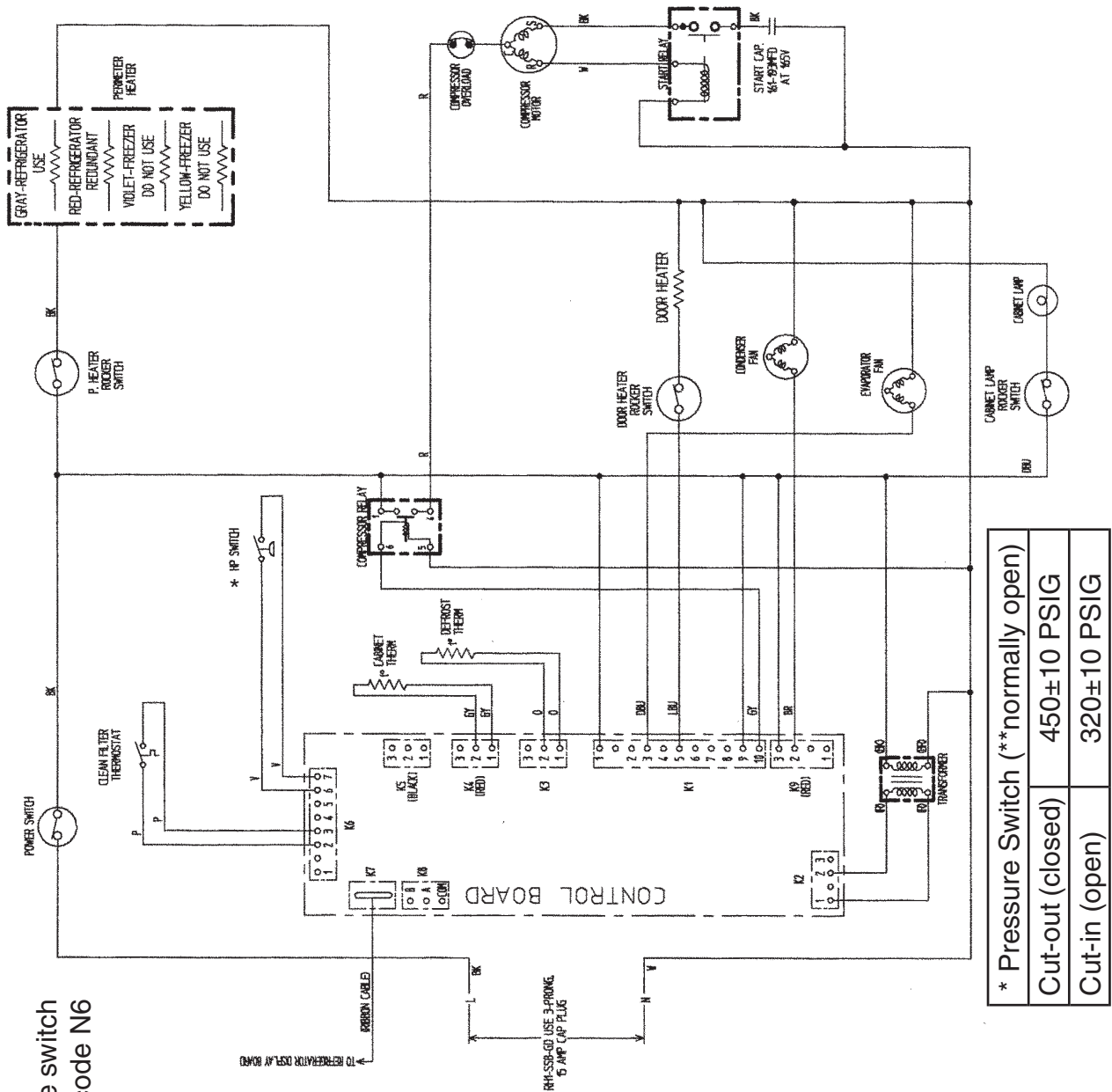
WIRE COLOR CODE	
BR	BROWN
W	WHITE
BK	BLACK
R	RED
O	ORANGE
GY	GRAY
P	PINK
DBU	DARK BLUE
LBUL	LIGHT BLUE
V	VIOLET

DIP SWITCH SETTING	
RH3-SSB(-HD)	
NO. 1	ON
NO. 2	OFF
NO. 3	OFF
NO. 4	OFF
NO. 5	OFF
NO. 6	OFF
NO. 7	OFF
NO. 8	OFF

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

D1. RH1-SSB-GD (auxiliary code N6 and earlier)

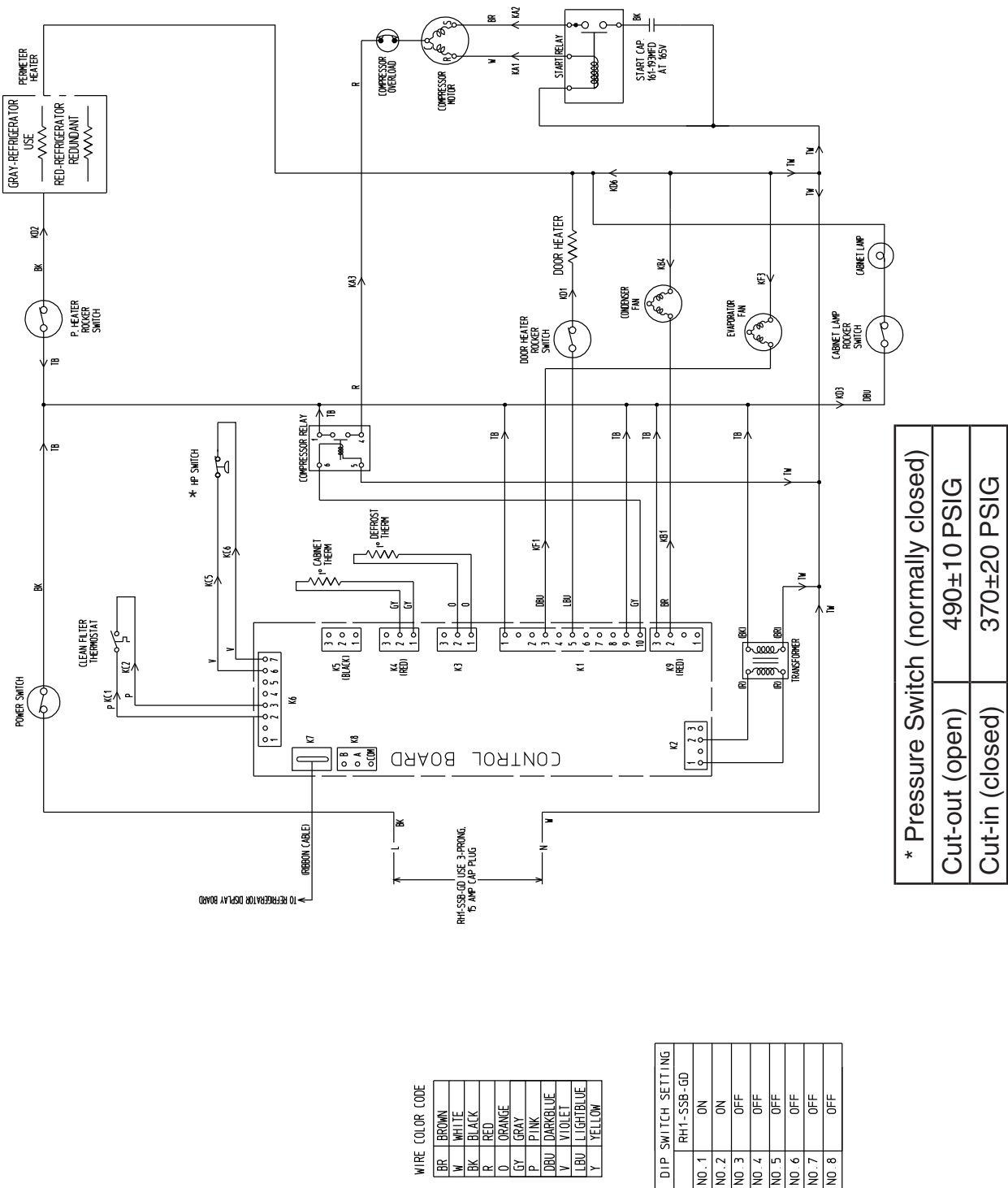


** Note: High pressure switch changed at auxiliary code N6 to normally closed. See "D2." for details.

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

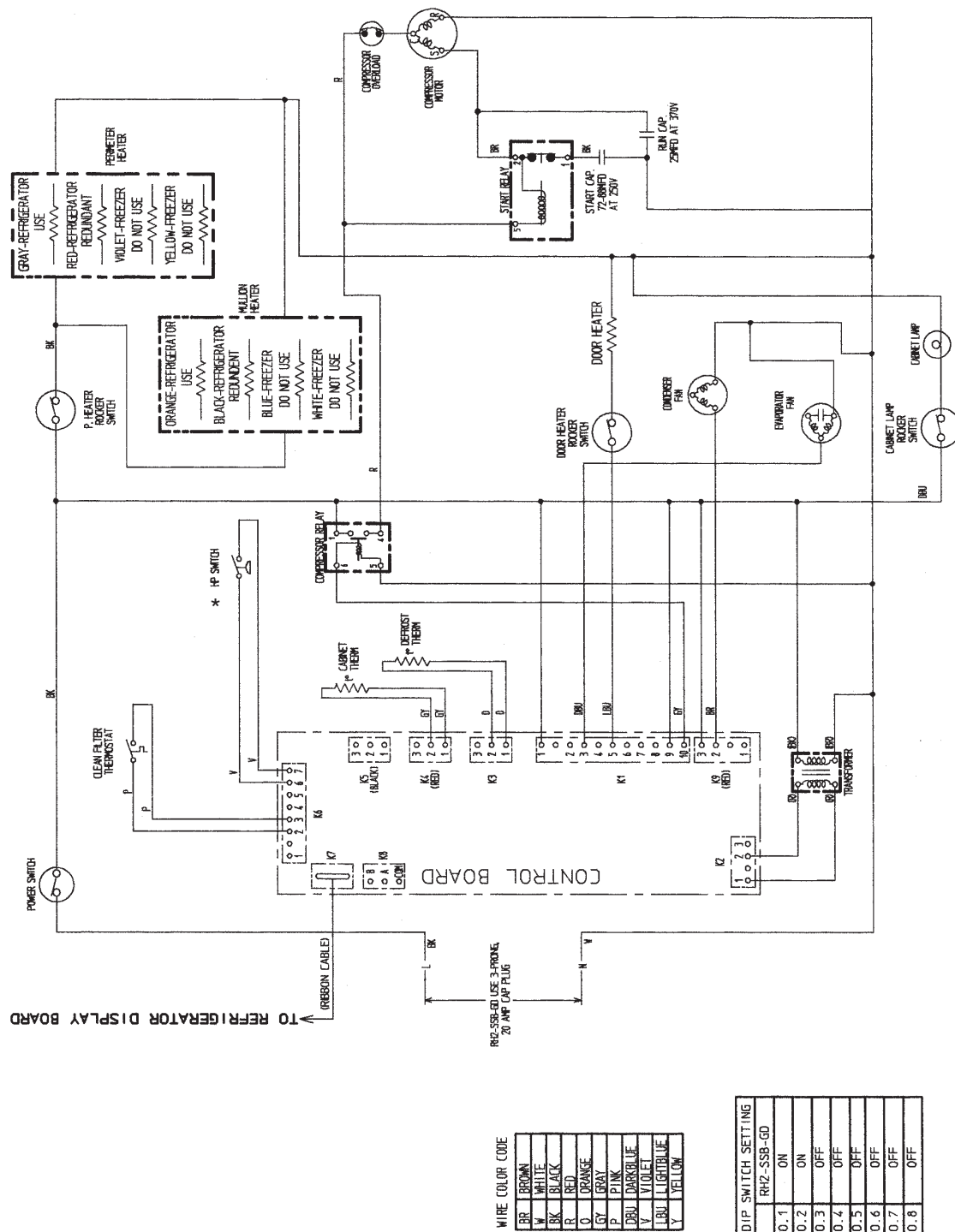
D2. RH1-SSB-GD (auxiliary code P5 and later)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

E1. RH2-SSB-GD (auxiliary code N6 and earlier)

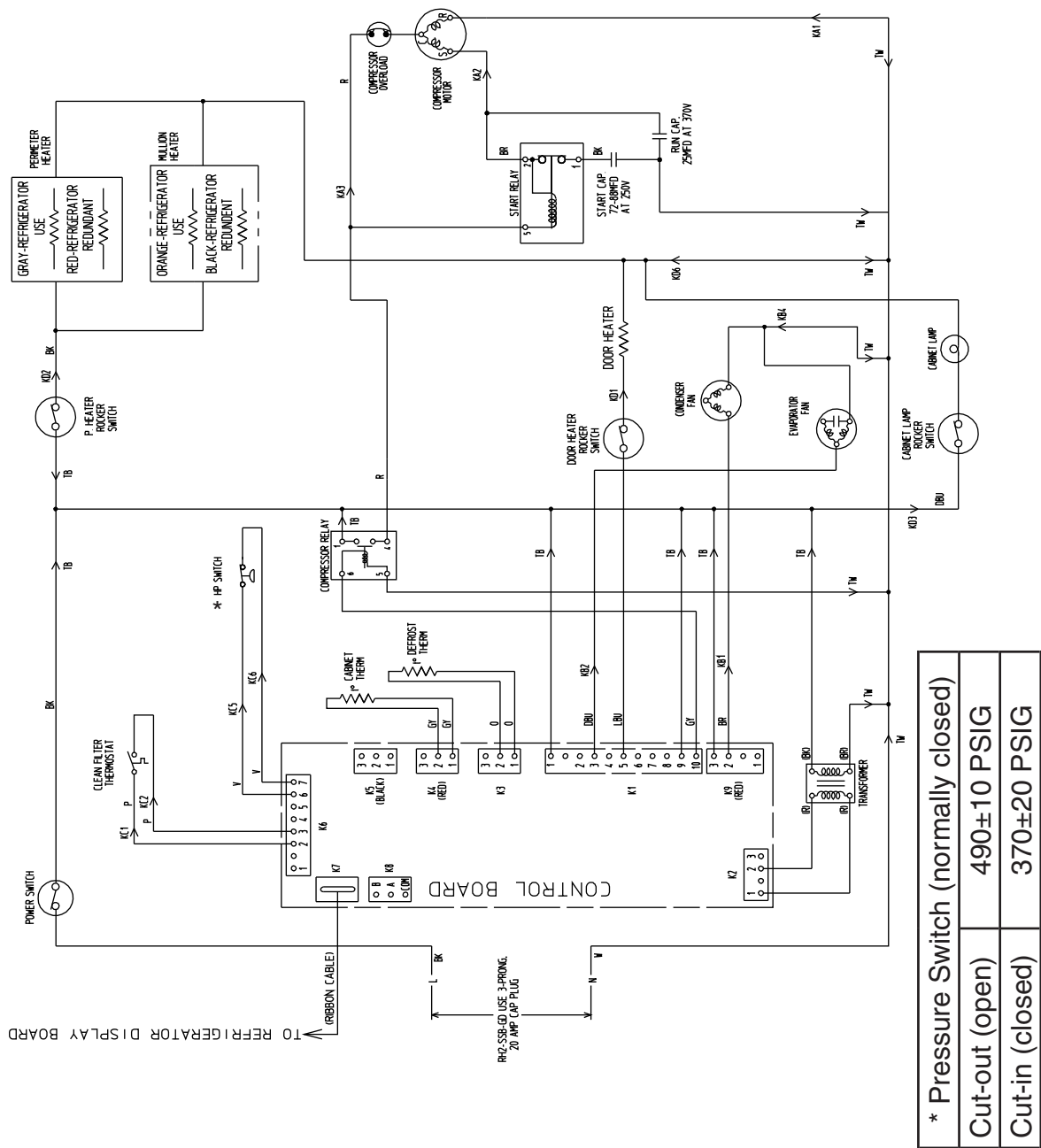


** Note: High pressure switch changed at auxiliary code M5 to normally closed. See "E2." for details.

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

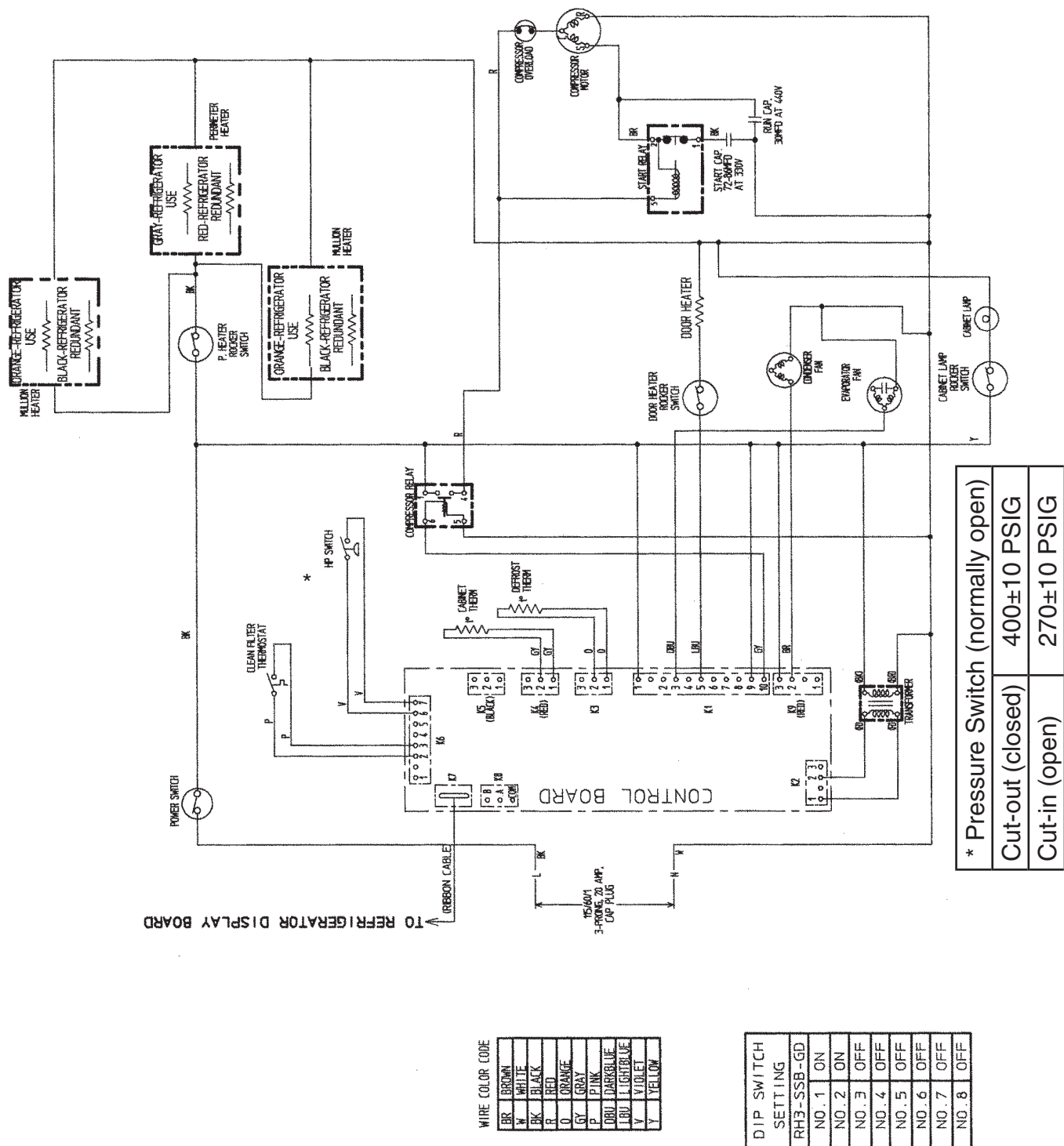
E2. RH2-SSB-GD (auxiliary code P5 and later)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

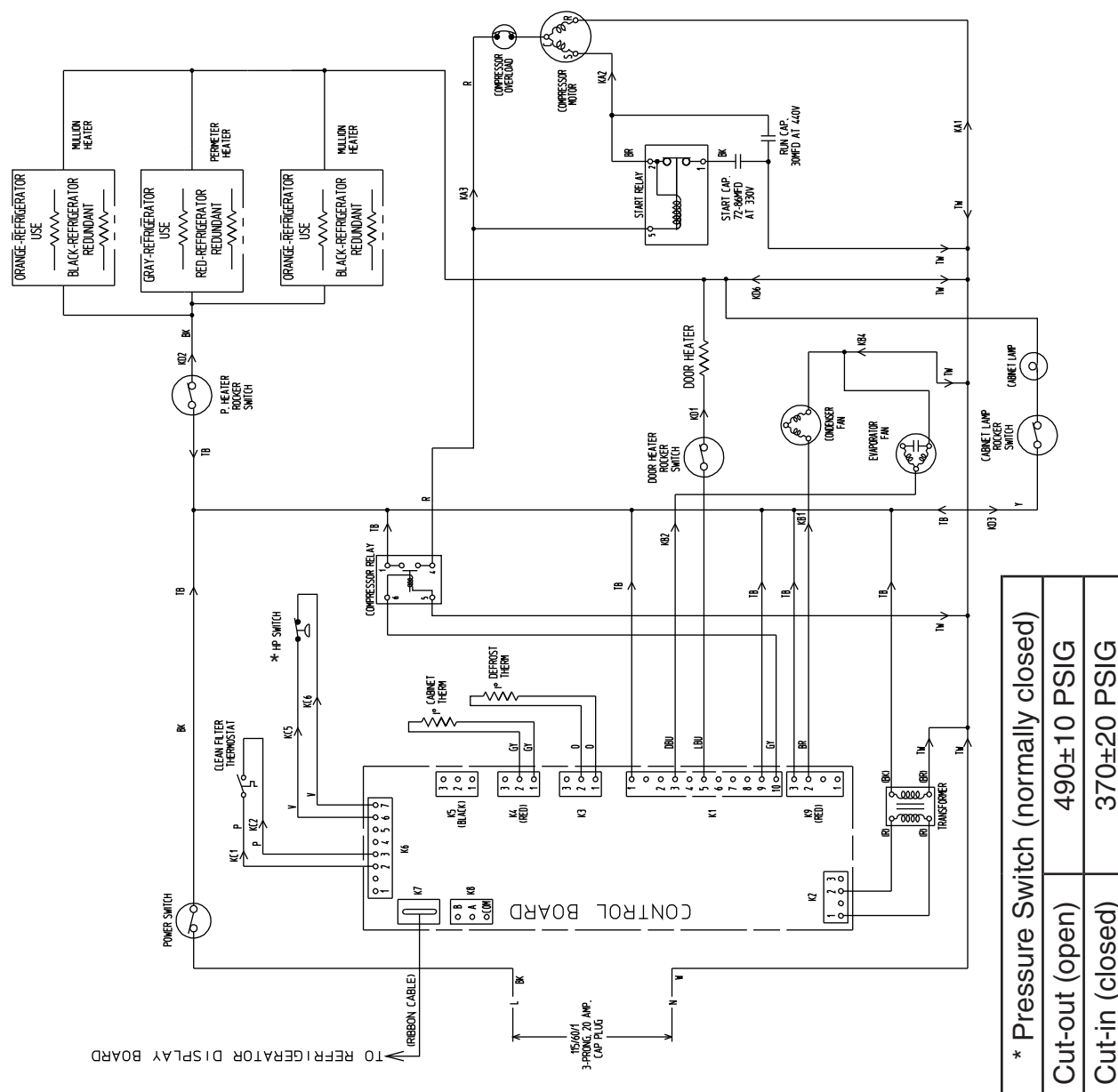
F1. RH3-SSB-GD (auxiliary code K0)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

F2. RH3-SSB-GD (auxiliary code N6 and later)



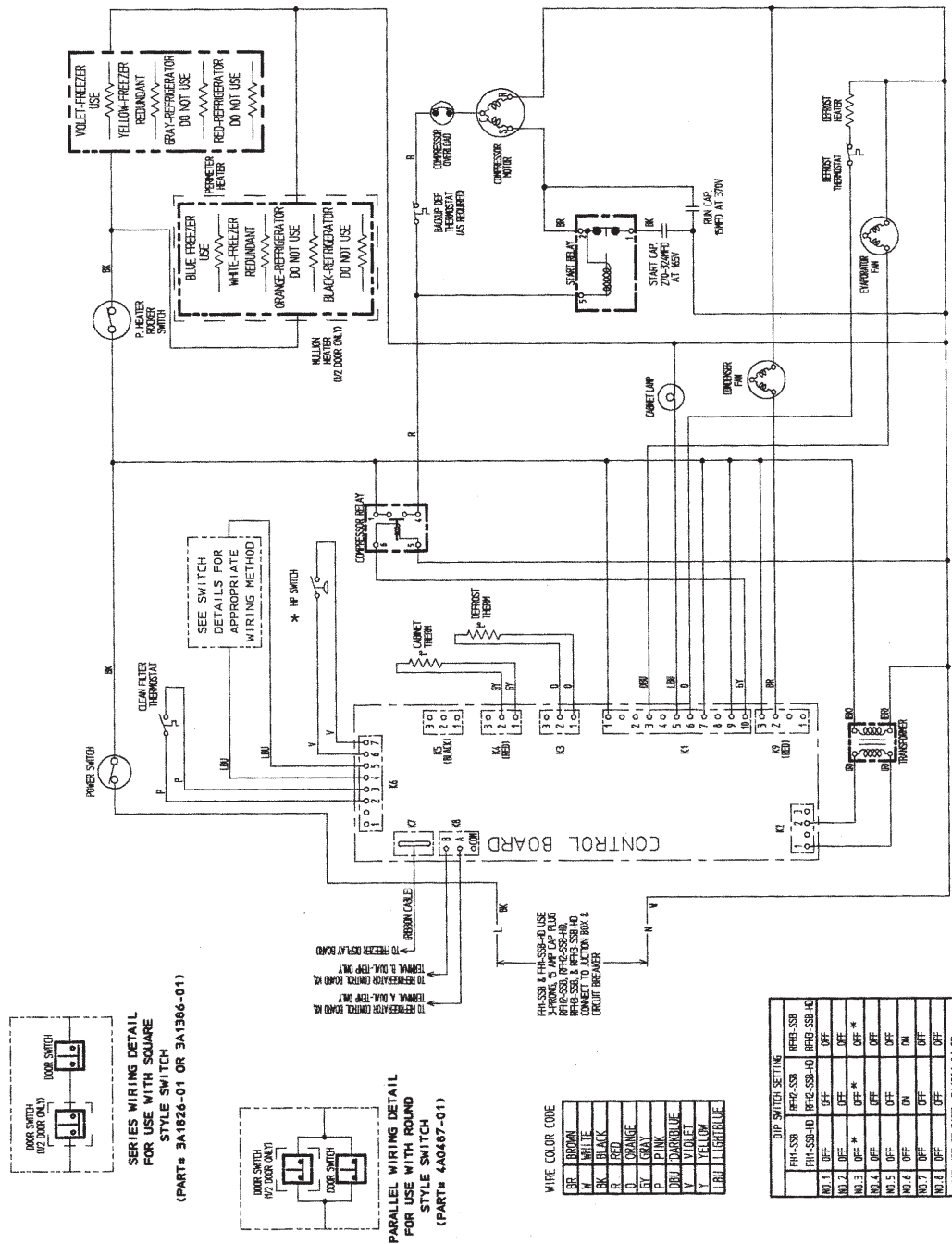
WIRE COLOR CODE	
BR	BROWN
W	WHITE
BK	BLACK
R	RED
O	ORANGE
GY	GRAY
P	PINK
DBU	DARKBLUE
LBV	LIGHTBLUE
V	VIOLET
Y	YELLOW

DIP SWITCH SETTING
RH3-SSB-GD
NO. 1 ON
NO. 2 ON
NO. 3 OFF
NO. 4 OFF
NO. 5 OFF
NO. 6 OFF
NO. 7 OFF
NO. 8 OFF

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

G1. FH1-SSB(-HD) (auxiliary code P5 and earlier)



* Pressure Switch (**normally open)

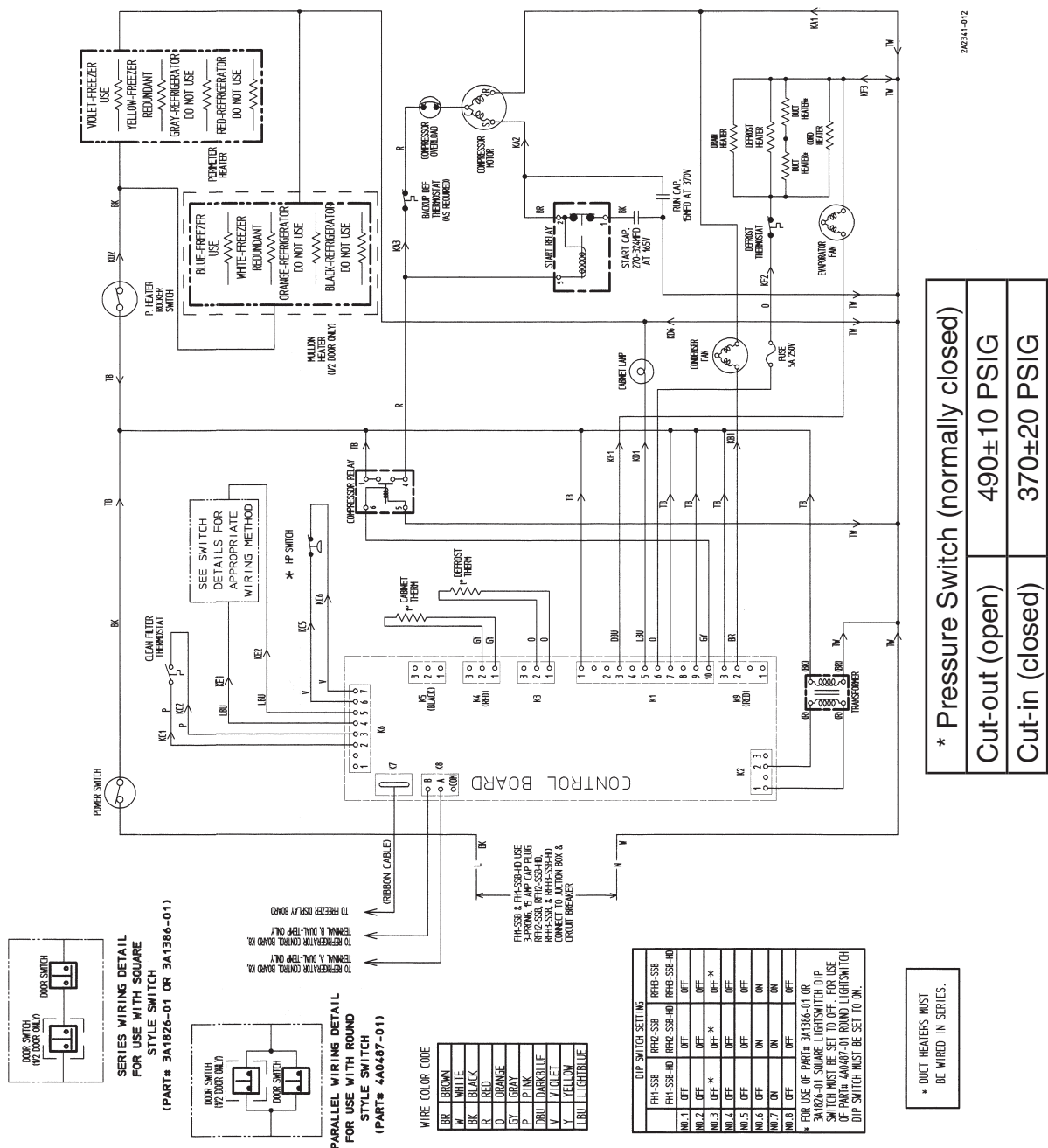
Cut-out (closed)	500±10 PSIG
Cut-in (open)	330±10 PSIG

**Note: High pressure switch changed at auxiliary code M6 to normally closed. See "G2." for details.

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

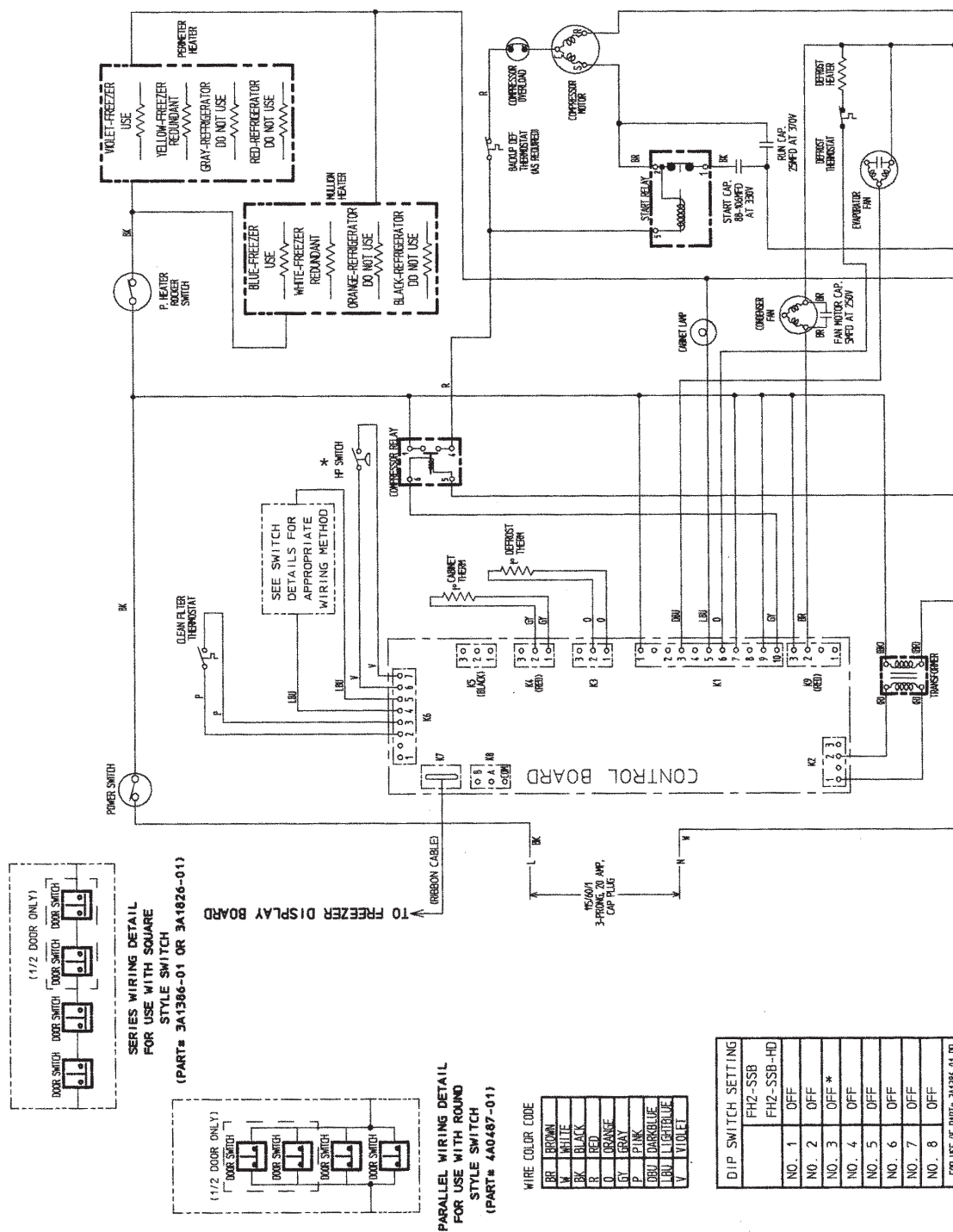
G2. FH1-SSB(-HD) (auxiliary code P6 and later)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

H1. FH2-SSB(-HD) (auxiliary code P5 and earlier)



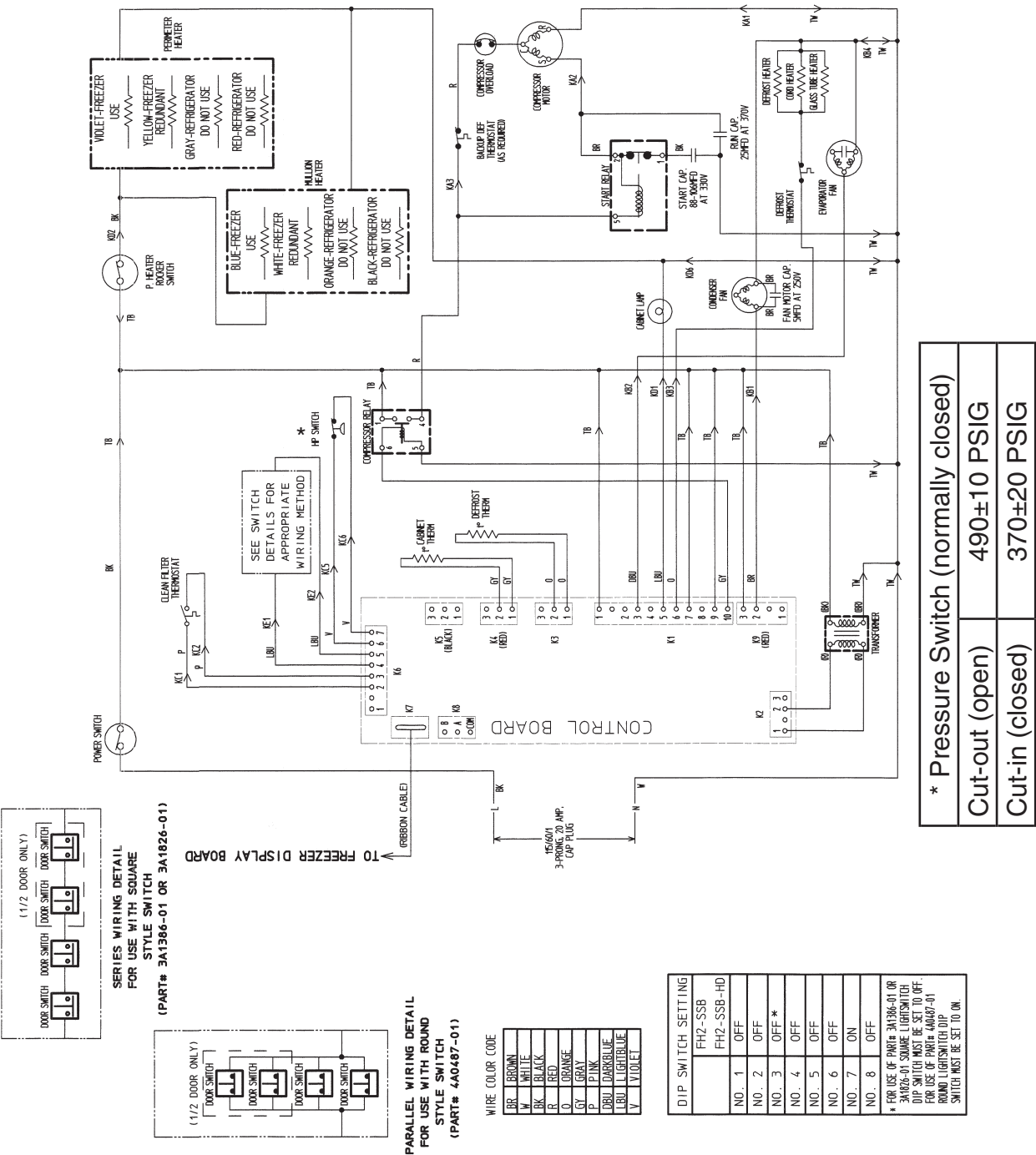
***Note: High pressure switch changed at auxiliary code L5 to normally closed. See "H2" for details.

* Pressure Switch (**normally open)		
	Aux code H0, J0	Aux code J1, J2
Cut-out (closed)	450±10 PSIG	500±10 PSIG
Cut-in (open)	320±10 PSIG	330±10 PSIG

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

H2. FH2-SSB(-HD) (auxiliary code P6 and later)

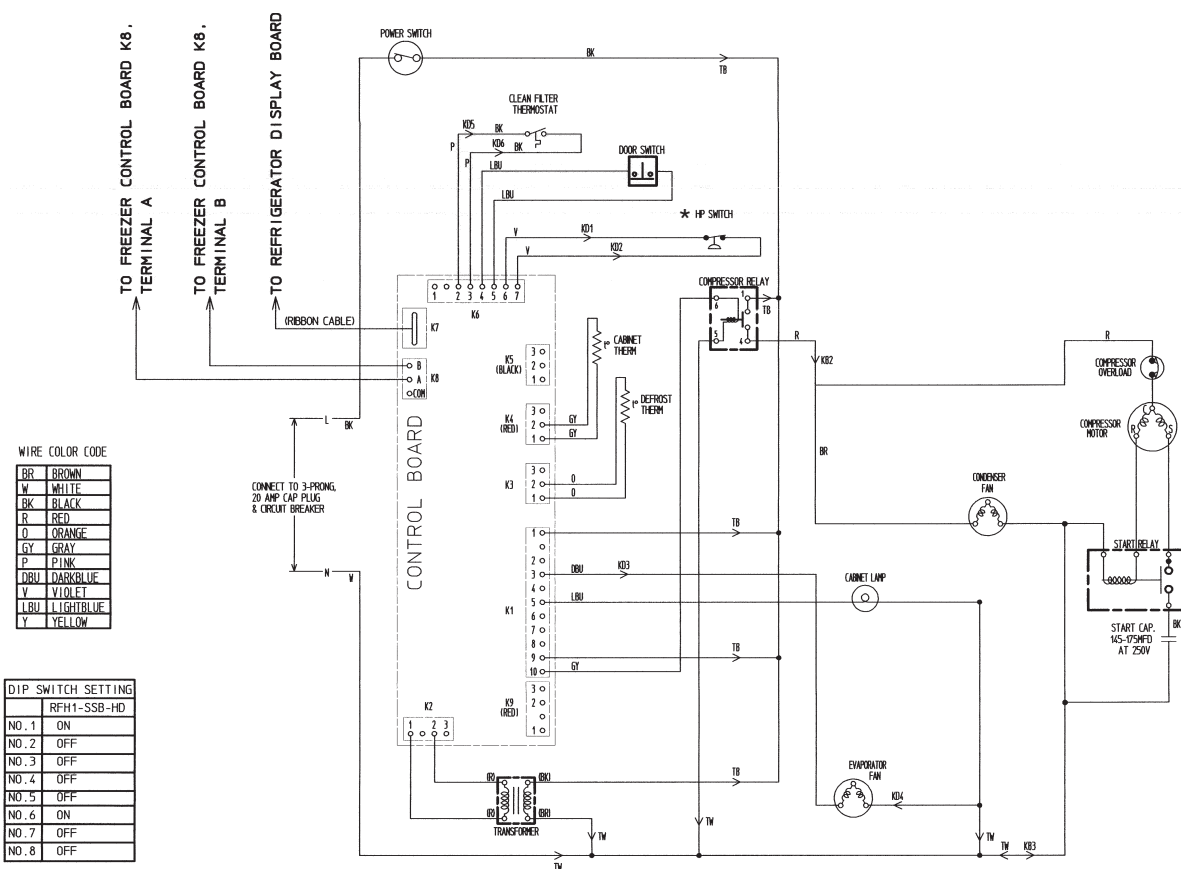


CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

I. RFH1-SSB-HD

1. Refrigerator

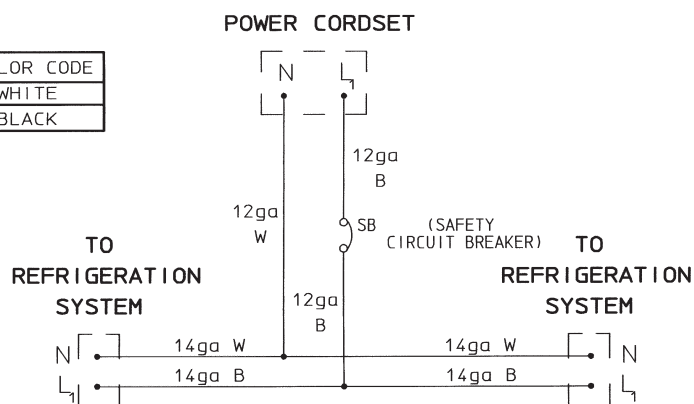


* Pressure Switch (normally closed)

Cut-out (open)	490± ¹⁰ PSIG
Cut-in (closed)	370± ²⁰ PSIG

JUNCTION BOX

WIRE	COLOR CODE
W	WHITE
B	BLACK

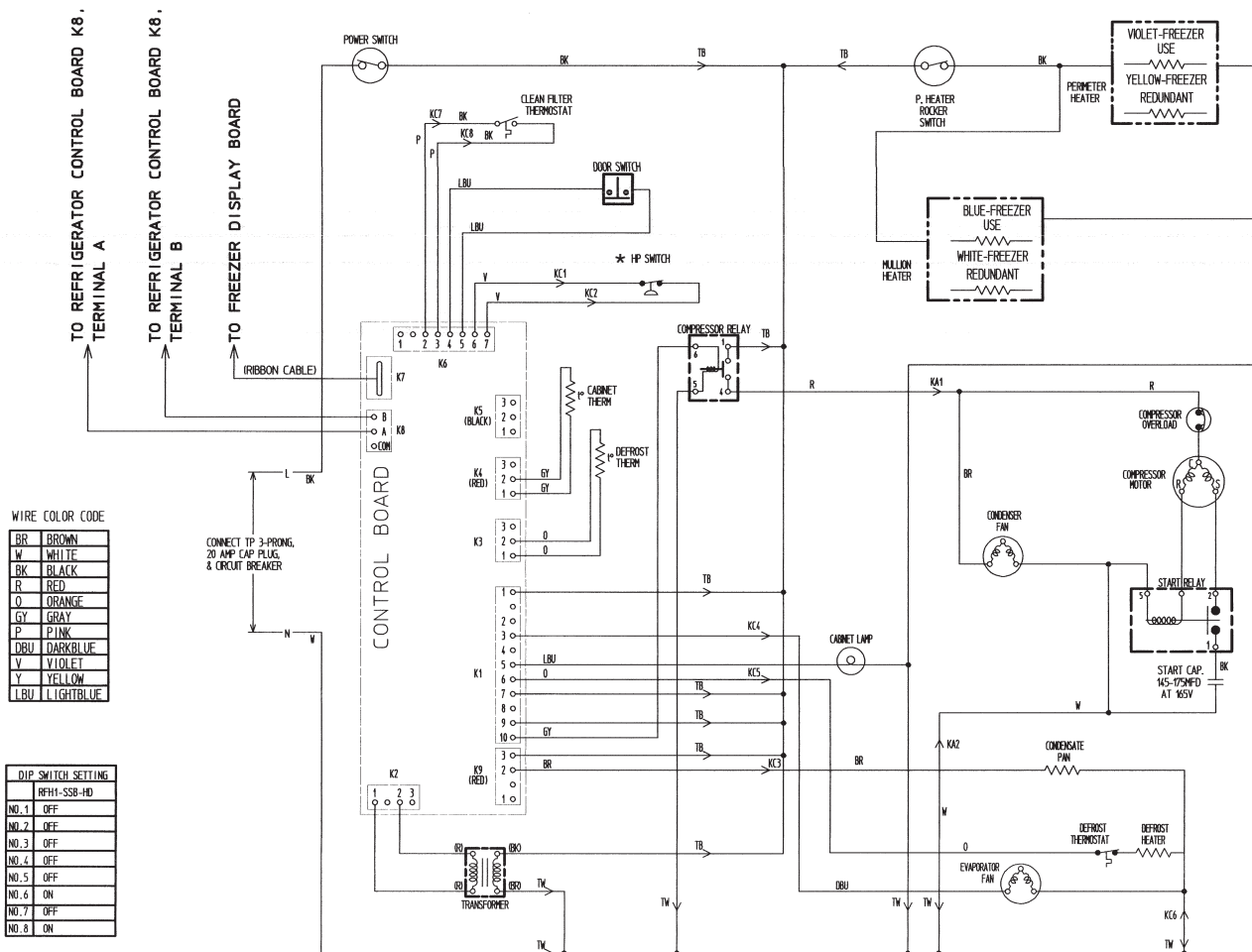


CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the top, front cover.

I. RFH1-SSB-HD

2a. Freezer (auxiliary code P5 and earlier)



* Pressure Switch (normally closed)

Cut-out (open)	490±10 PSIG
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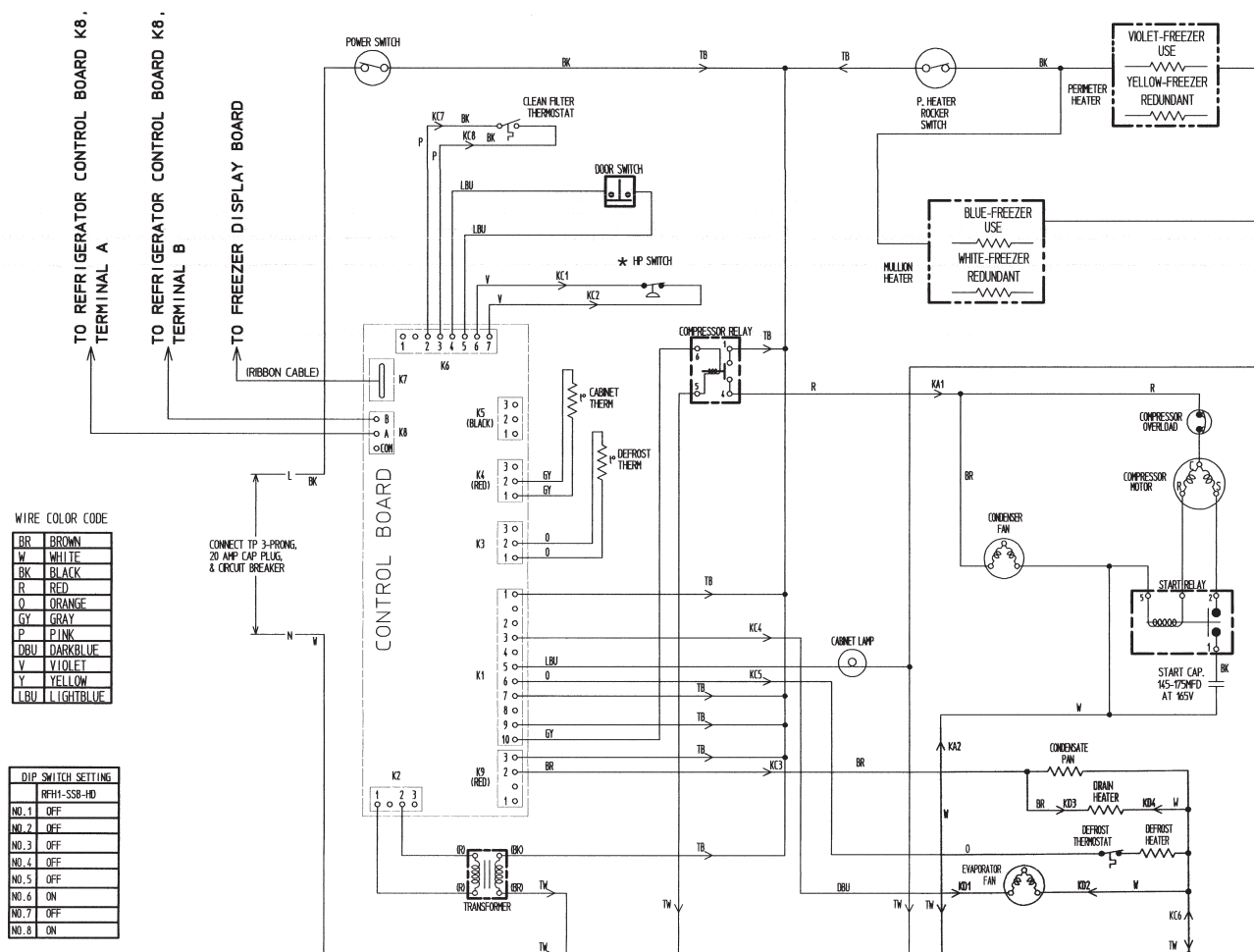
Cut-in (closed)	370±20 PSIG
-----------------	-------------

CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the top, front cover.

I. RFH1-SSB-HD

2b. Freezer (auxiliary code Q5 and later)

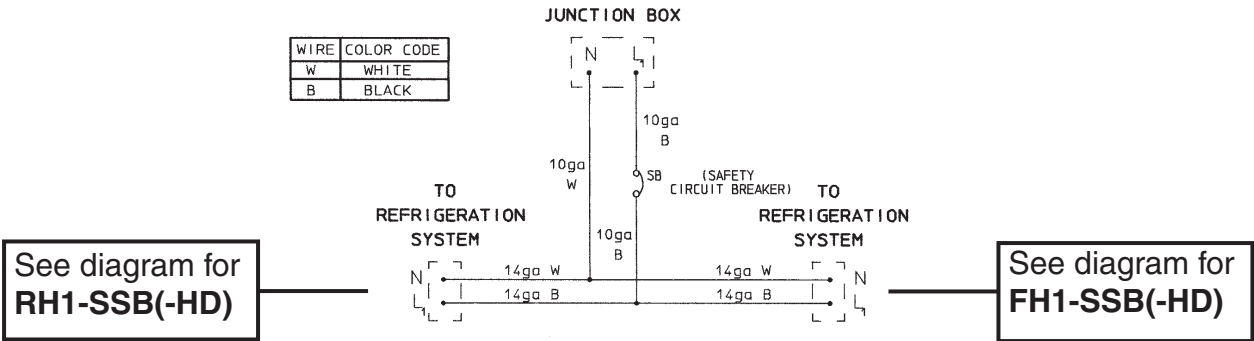


* Pressure Switch (normally open)	
Cut-out (open)	490±10 PSIG
Cut-in (closed)	370±20 PSIG

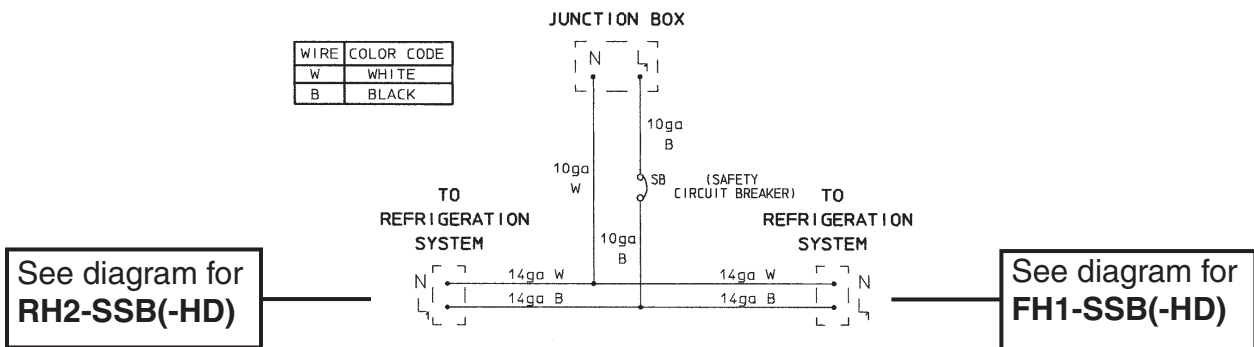
CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the top, front cover.

J. RFH2-SSB(-HD)



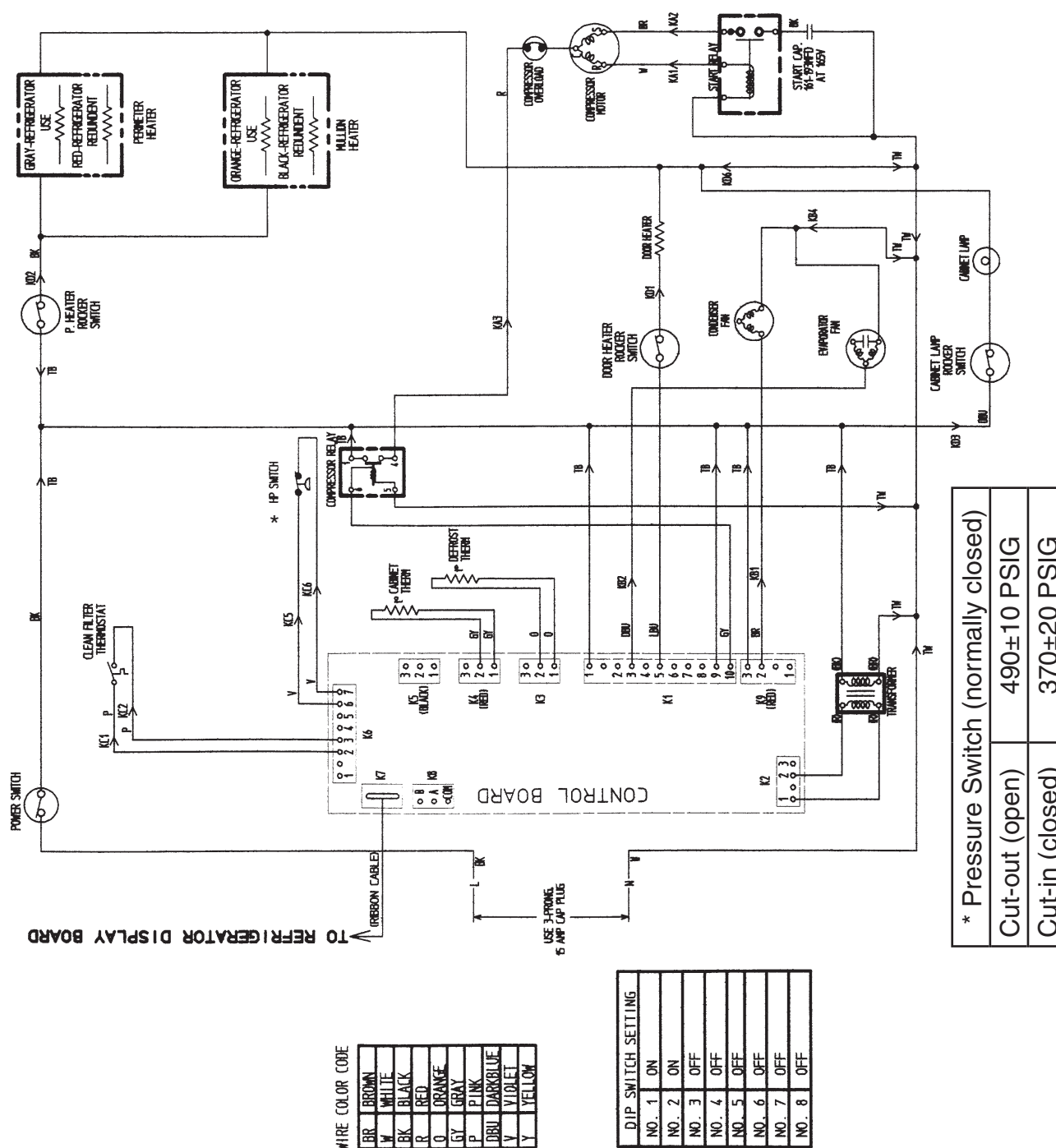
K. RFH3-SSB(-HD)



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

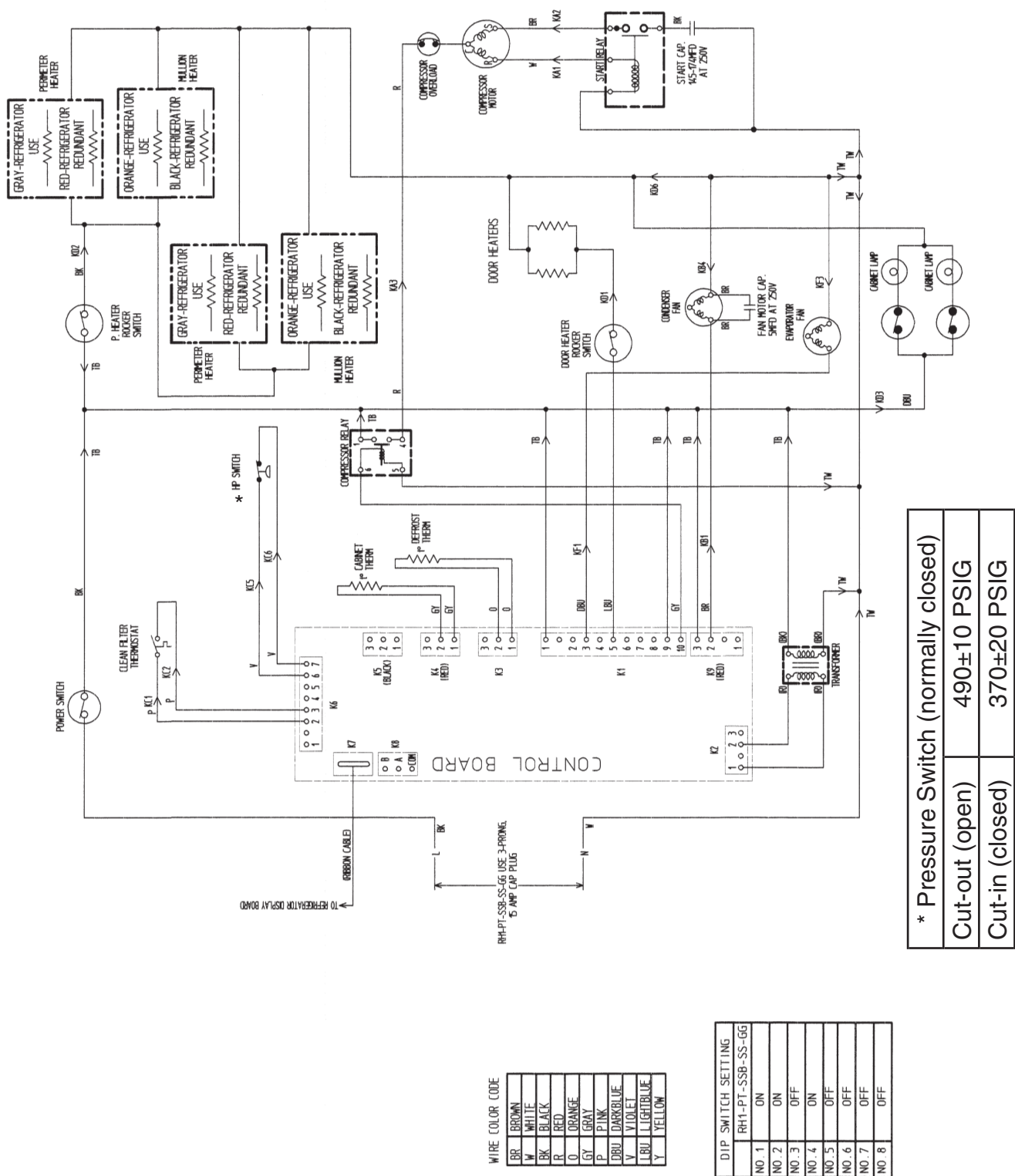
L. RIR2-SSB51-02



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

M. PTR1SSB-xxxx, PTR1-SSB01-10



CAUTION

Before diagnosing and/or servicing the unit, confirm that you have the proper wiring diagram by checking against the diagram located on the evaporator case cover.

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